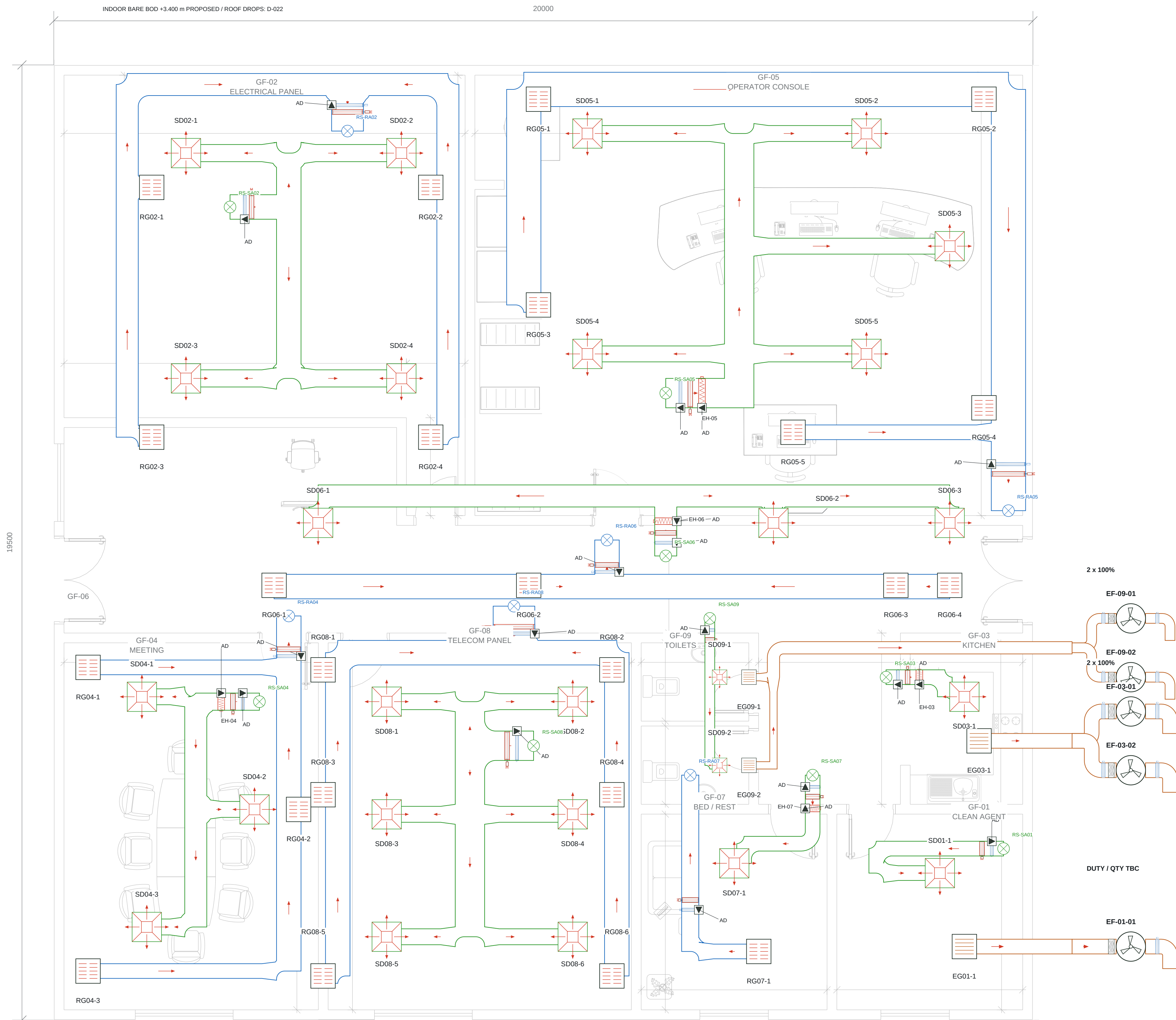


## MAIN DUCT AND RETURN - COORDINATED DUCT PLAN

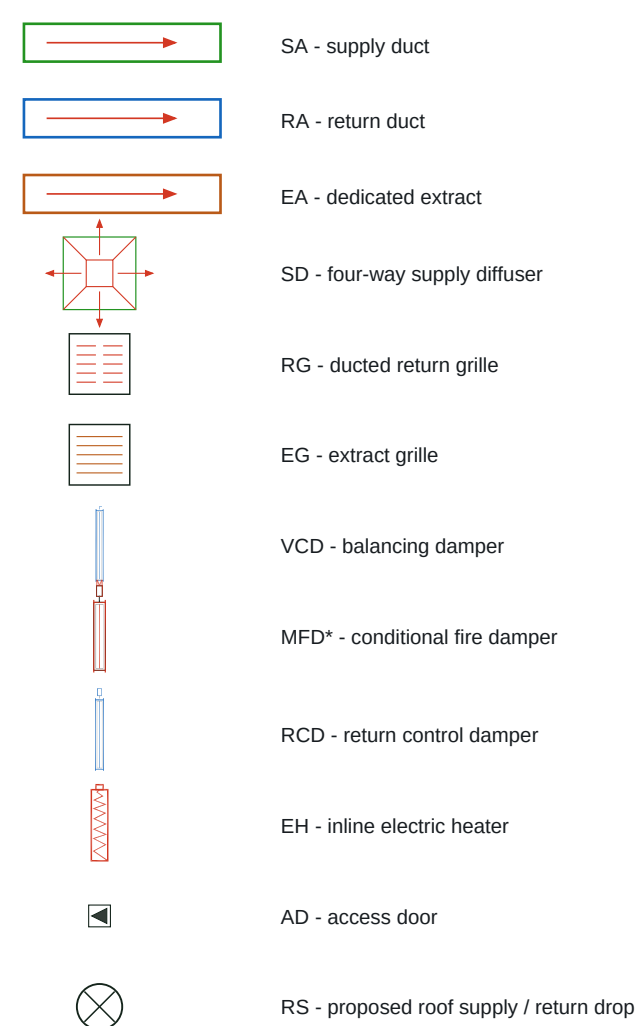
REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033..035



### ROOM AIRFLOW SCHEDULE

| GF | Room             | SA L/s | RA L/s* | EA L/s* |
|----|------------------|--------|---------|---------|
| 02 | ELECTRICAL PANEL | 664.8  | 605.5   | -       |
| 05 | OPERATOR CONSOLE | 777.7  | 718.4   | -       |
| 04 | MEETING          | 449.2  | 389.9   | -       |
| 08 | TELECOM PANEL    | 884.6  | 825.3   | -       |
| 06 | CORRIDOR         | 418.2  | 358.9   | -       |
| 07 | BED / REST       | 168.2  | 108.9   | -       |
| 09 | TOILETS          | 46.2   | -       | 64.9    |
| 03 | KITCHEN          | 147.6  | -       | 147.6   |
| 01 | CLEAN AGENT      | 141.3  | -       | TBC     |

### DUCT / SYMBOL LEGEND



Terminal aliases: SD02-1 = SD-02-01; RG / EG follow the same format. Full tags and quantities are in the registers.

Roof-main OPTION: 15 independent room drops RS-SA / RS-RA. No indoor service crossings. Finished ceiling +3.300 m; proposed bare BOD +3.400 m. Envelope: D-020; roof/drop holds: D-022.

\* Return quantities are assumed balances. Extract references and +50 / +25 Pa targets require final airflow and pressure verification

Units: mm / L/s / m/s. Terminal faces and clear duct widths are shown. Duct levels, selected fitting development and manufacturer clearances remain pending. No shared return from kitchen, toilet or clean-agent room.

## TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

## ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

## COORDINATED DUCT PLAN

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-001

SCALE: 1:50 AT A1

REVISION

02

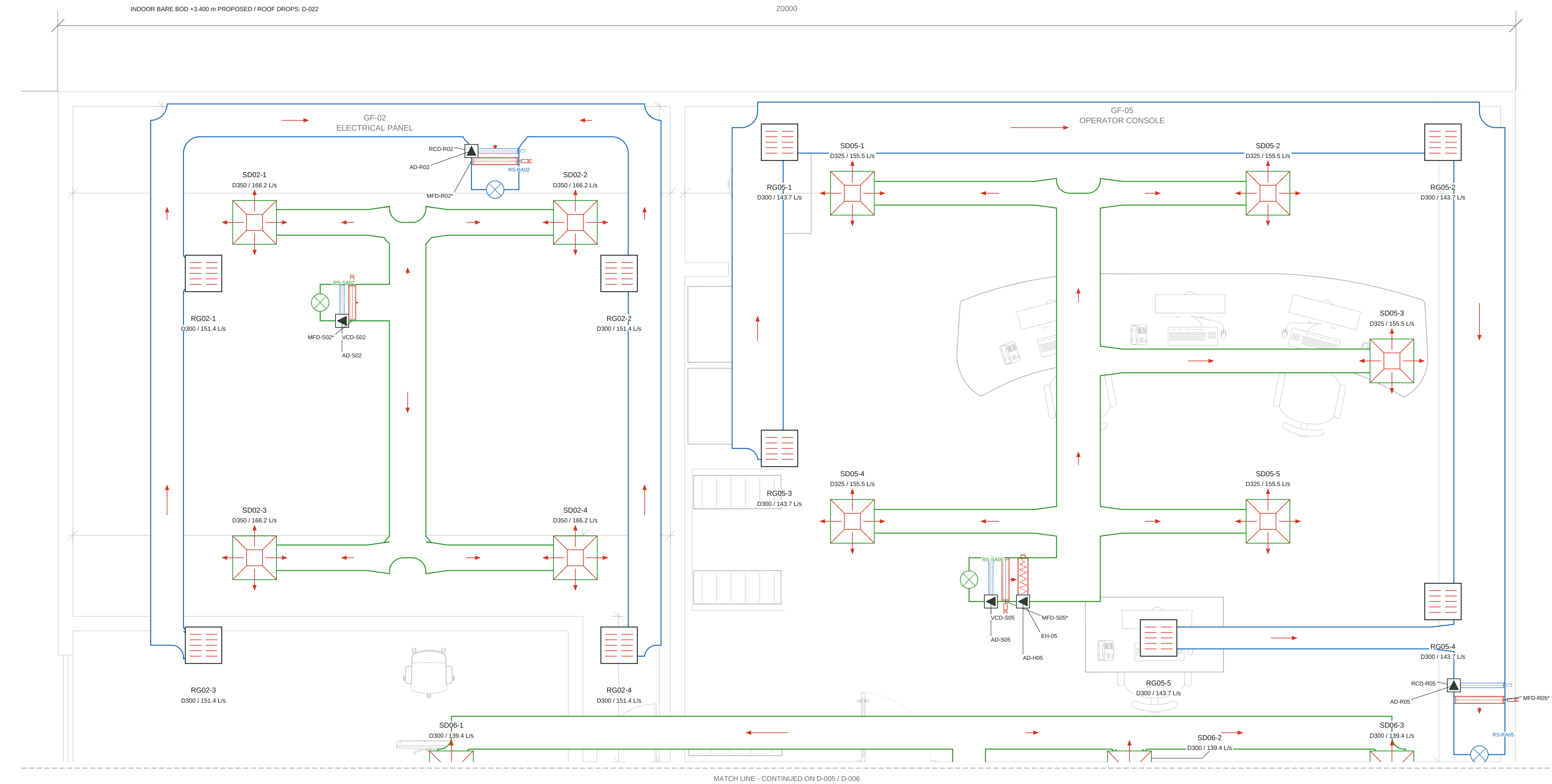
08 OCT 2026

SHEET 1 OF 50



MAIN DUCT AND RETURN - NORTH ROOMS - DUCT ENLARGEMENT

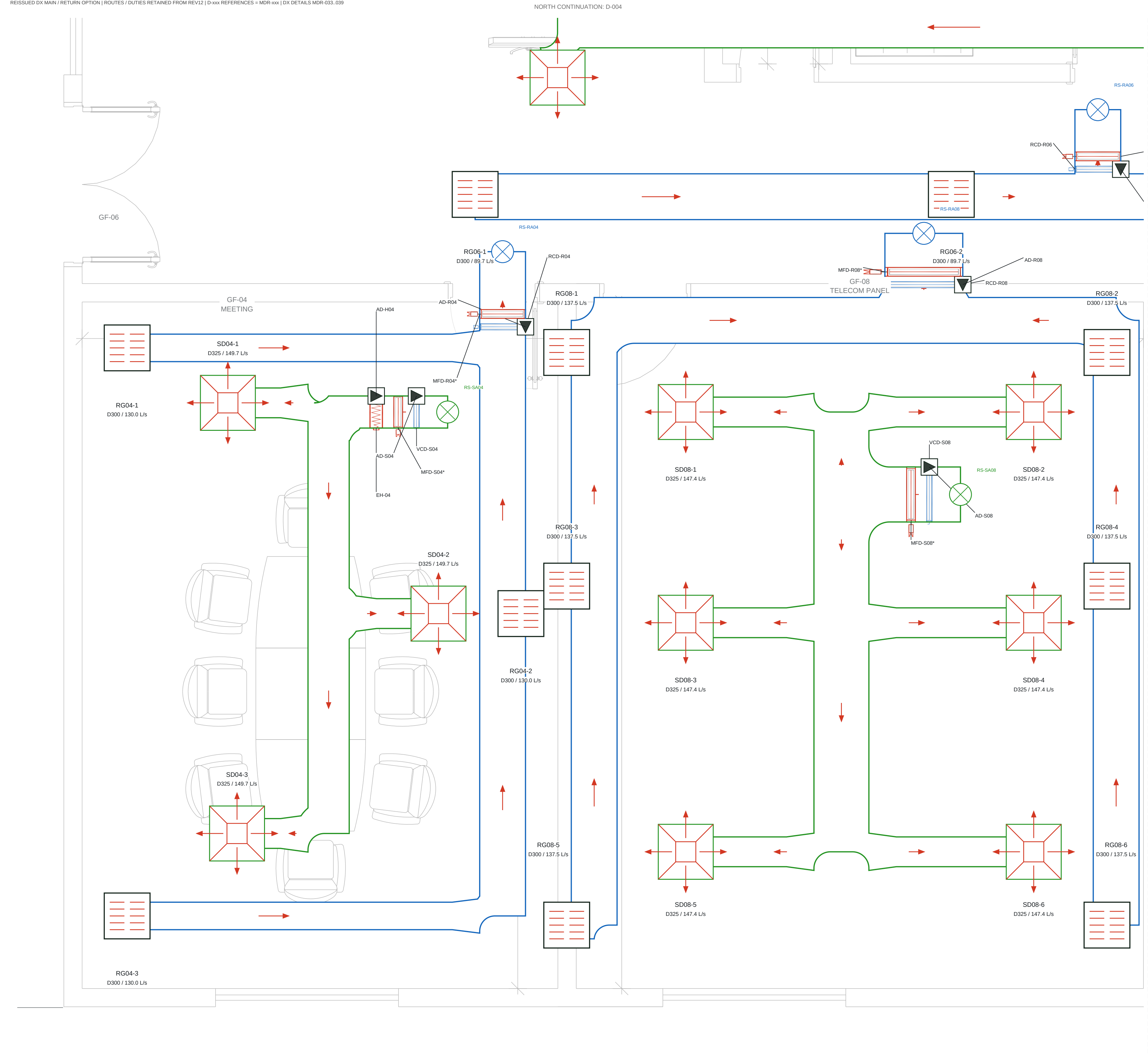
REISSUED DX MAIN / RETURN OPTION / ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033\_039





MAIN DUCT AND RETURN - SOUTH-WEST ROOMS - DUCT ENLARGEMENT

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033\_039



ROOM BRANCH / TERMINAL DATA

GF-04 MEETING

SA-B04 350x400 mm | 449.20 L/s | 3.21 m/s

RA-B04 500x350 mm | 389.86 L/s | 2.23 m/s

SD: 3 x 149.73 L/s; neck D325; face 600x600

RG: 3 x 129.95 L/s; neck D300; face 500x500

Room header: branch size. Terminal runoff: listed neck.

GF-08 TELECOM PANEL

SA-B08 600x400 mm | 884.60 L/s | 3.69 m/s

RA-B08 850x400 mm | 825.26 L/s | 2.43 m/s

SD: 6 x 147.43 L/s; neck D325; face 600x600

RG: 6 x 137.54 L/s; neck D300; face 500x500

Room header: branch size. Terminal runoff: listed neck.

GF-06 CORRIDOR

SA-B06 450x300 mm | 418.20 L/s | 3.10 m/s

RA-B06 500x400 mm | 358.86 L/s | 1.79 m/s

SD: 3 x 139.40 L/s; neck D300; face 600x600

RG: 4 x 89.72 L/s; neck D300; face 500x500

Room header: branch size. Terminal runoff: listed neck.

DUCT / SYMBOL LEGEND

- SA - supply duct
- RA - return duct
- EA - dedicated extract
- SD - four-way supply diffuser
- RG - ducted return grille
- EG - extract grille
- VCD - balancing damper
- MFD\* - conditional fire damper
- RCD - return control damper
- EH - inline electric heater
- AD - access door
- RS - proposed roof supply / return drop

Red arrows show air direction; green outlines are supply and blue outlines are return. Room drops refer to D-022. No indoor service crossovers. Same-service junctions are continuous open tees; headroom envelope: D-020.

GF-04 meeting; GF-08 telecom; GF-06 corridor. Room targets +50 Pa relative to outdoors; actual pressure verification remains pending. Full flow / velocity / horizontal route data is listed on D-010.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

SOUTH-WEST ROOMS - DUCT ENLARGEMENT

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-005

SCALE: 1:25 AT A1

REVISION

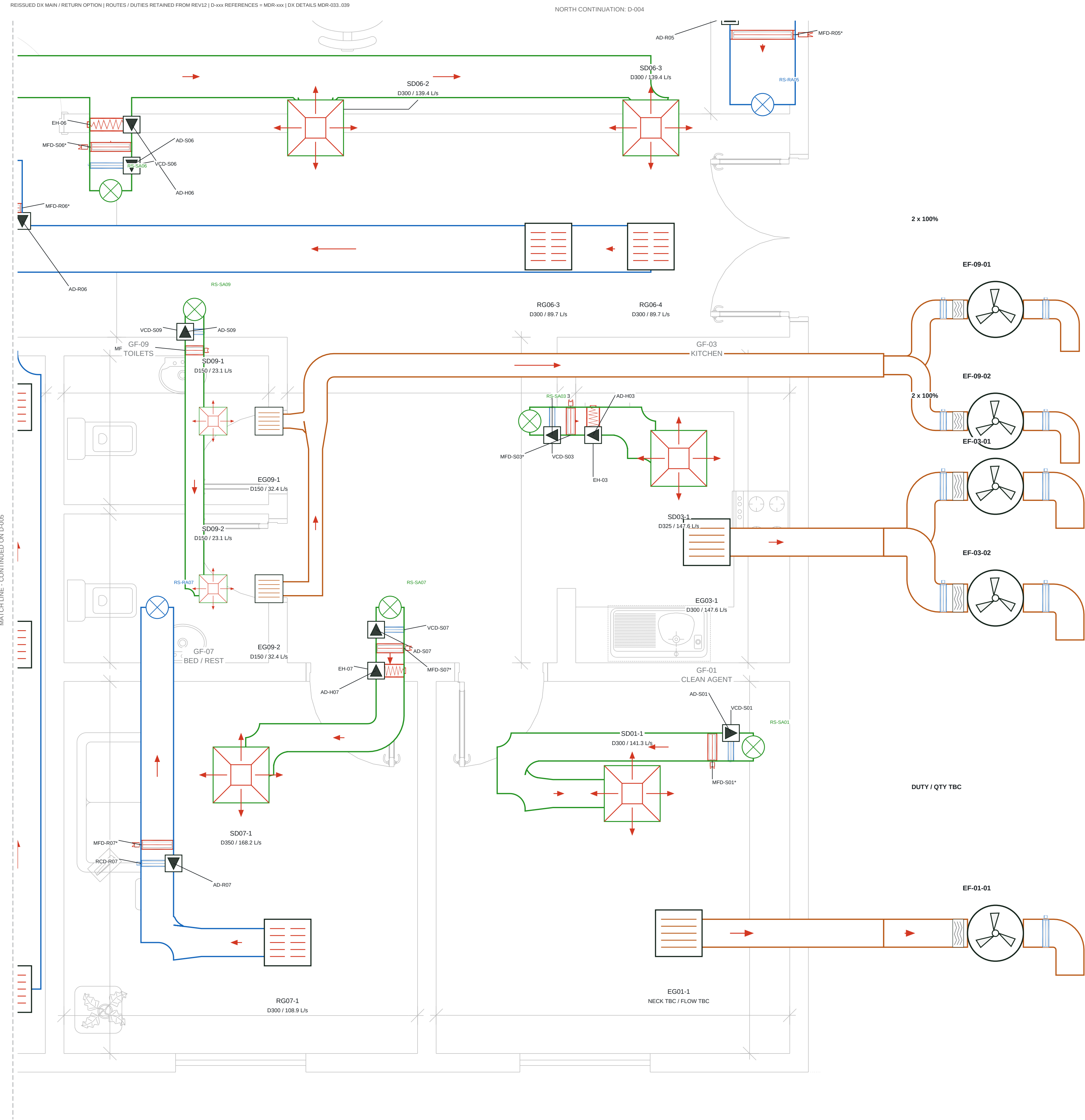
02

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SHEET 5 OF 50



MAIN DUCT AND RETURN - SOUTH-EAST ROOMS - DUCT ENLARGEMENT



ROOM BRANCH / TERMINAL DATA

GF-09 TOILETS

SA-B09 200x150 mm | 46.20 L/s | 1.54 m/s

EA-B09 250x200 mm | 64.89 L/s | 1.30 m/s

SD: 2 x 23.10 L/s; neck D150; face 300x300

Dedicated extract; no central return

Room header: branch size. Terminal runout: listed neck.

GF-07 BED / REST

SA-B07 300x250 mm | 168.20 L/s | 2.24 m/s

RA-B07 350x250 mm | 108.86 L/s | 1.24 m/s

SD: 1 x 168.20 L/s; neck D350; face 600x600

RG: 1 x 108.86 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

GF-03 KITCHEN

SA-B03 300x200 mm | 147.60 L/s | 2.46 m/s

EA-B03 300x200 mm | 147.60 L/s | 2.46 m/s

SD: 1 x 147.60 L/s; neck D325; face 600x600

Dedicated extract; no central return

Room header: branch size. Terminal runout: listed neck.

GF-01 CLEAN AGENT

SA-B01 300x200 mm | 141.30 L/s | 2.35 m/s

SD: 1 x 141.30 L/s; neck D300; face 600x600

EA / grille duty and pickup level: TBC

Room header: branch size. Terminal runout: listed neck.

EXTRACT ARRANGEMENTS

EF-09-01/02: toilets, 2 x 100%. EF-03-01/02: kitchen, 2 x 100%. EF-01-01: one CAG fan marker shown; final quantity, airflow and pickup elevation require confirmation.

Kitchen, toilet and CAG targets: +25 Pa to outdoors. Resting room: +50 Pa, with return air. Minimum ventilation: 10 ACH kitchen / toilet; 5 ACH CAG.

Toilet HAP supply is 46.2 L/s; minimum outdoor-air duty is not established by that mixed-air supply. Resolve the fresh-air duty before final HAP and pressure balance.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

SOUTH-EAST ROOMS - DUCT ENLARGEMENT

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-006

SCALE: 1:25 AT A1

REVISION

02

08 OCT 2026

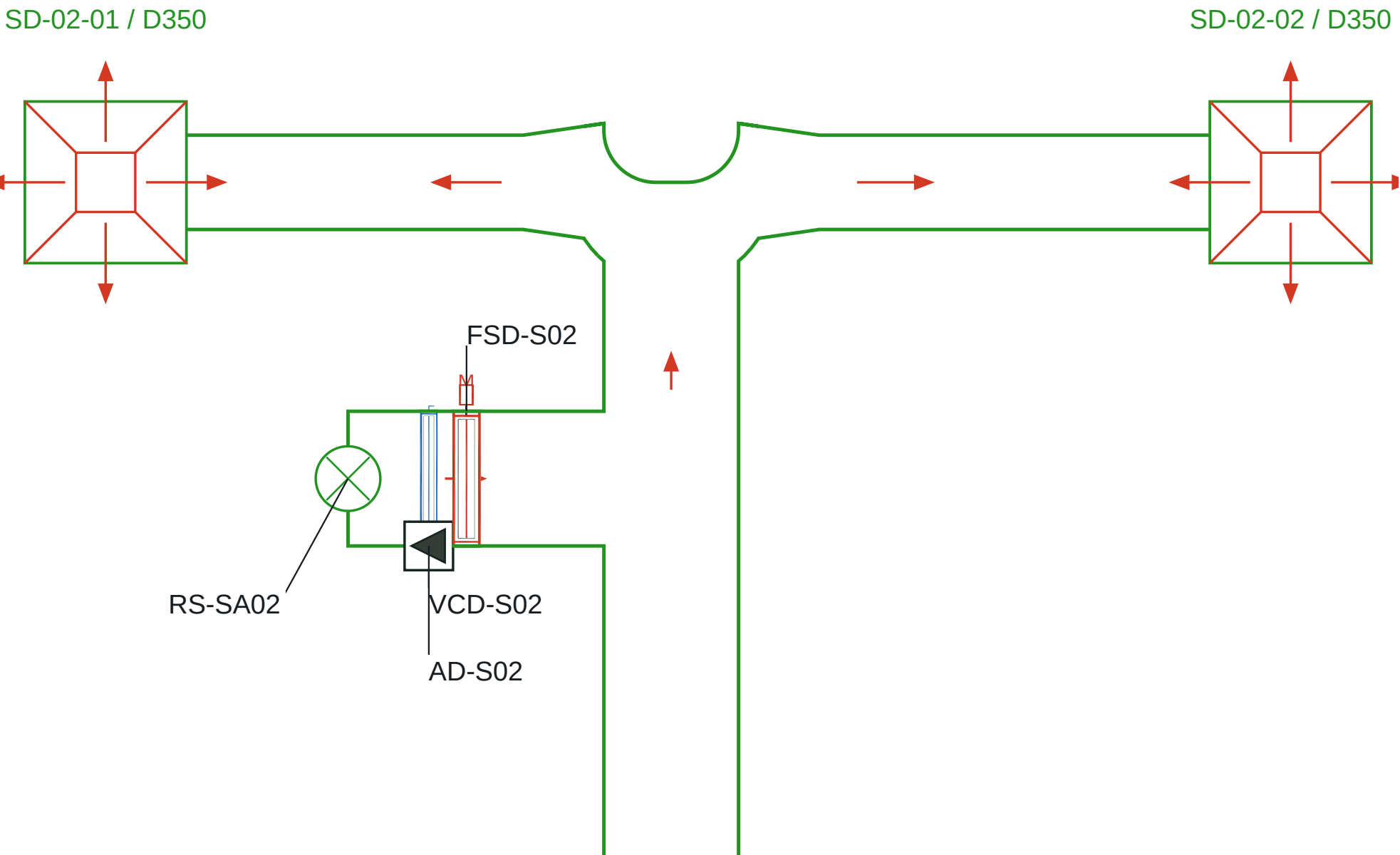
SHEET 6 OF 50



MAIN DUCT AND RETURN - ACTUAL DUCT JUNCTIONS - ENLARGED PLAN DETAILS

REISSUED DX MAIN / RETURN OPTION / ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033\_039

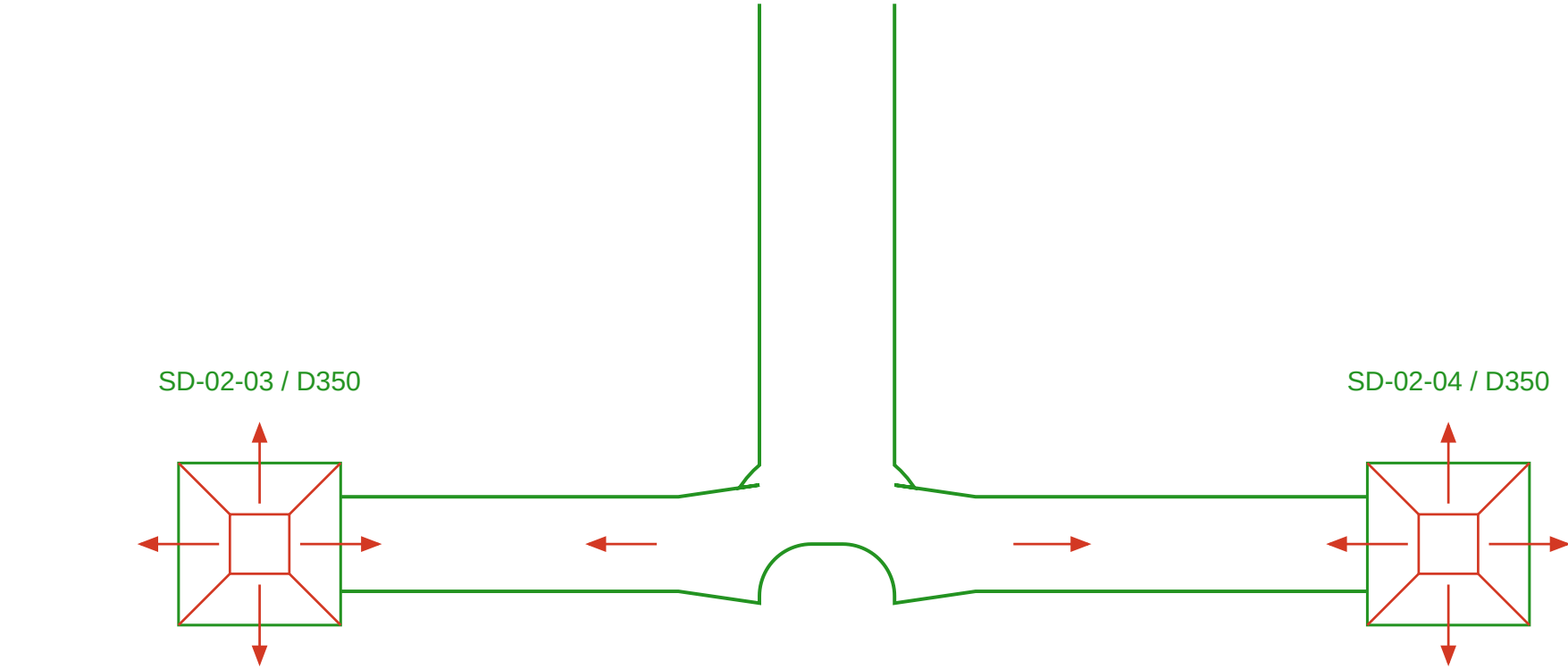
A GF-02 ROOF DROP / BRANCH / HEADER LEGS / 1:20



Actual Rev11 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct\_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

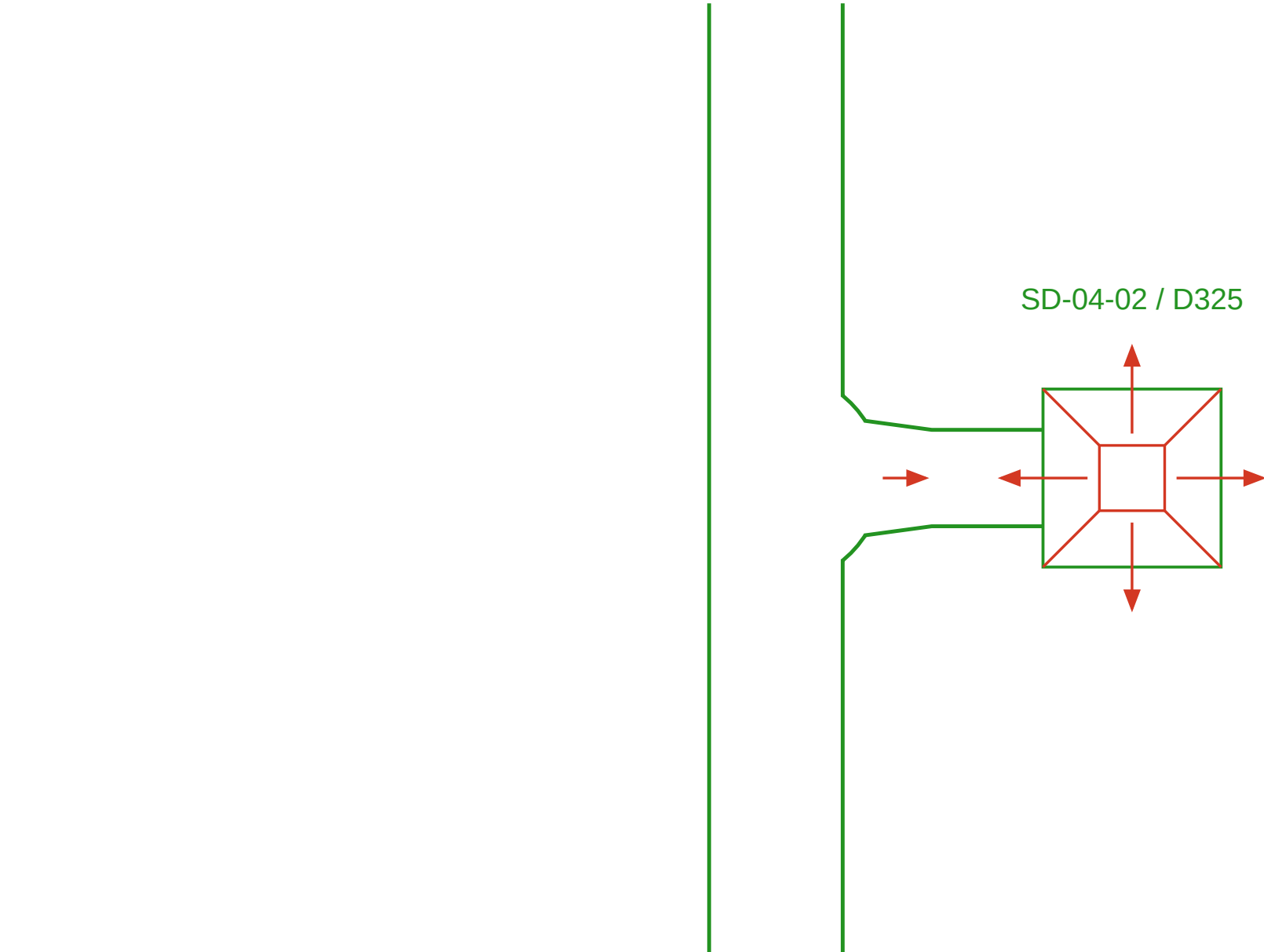
B GF-02 END TEE / TWO INDIVIDUAL NECKS / 1:25



Actual Rev11 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct\_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

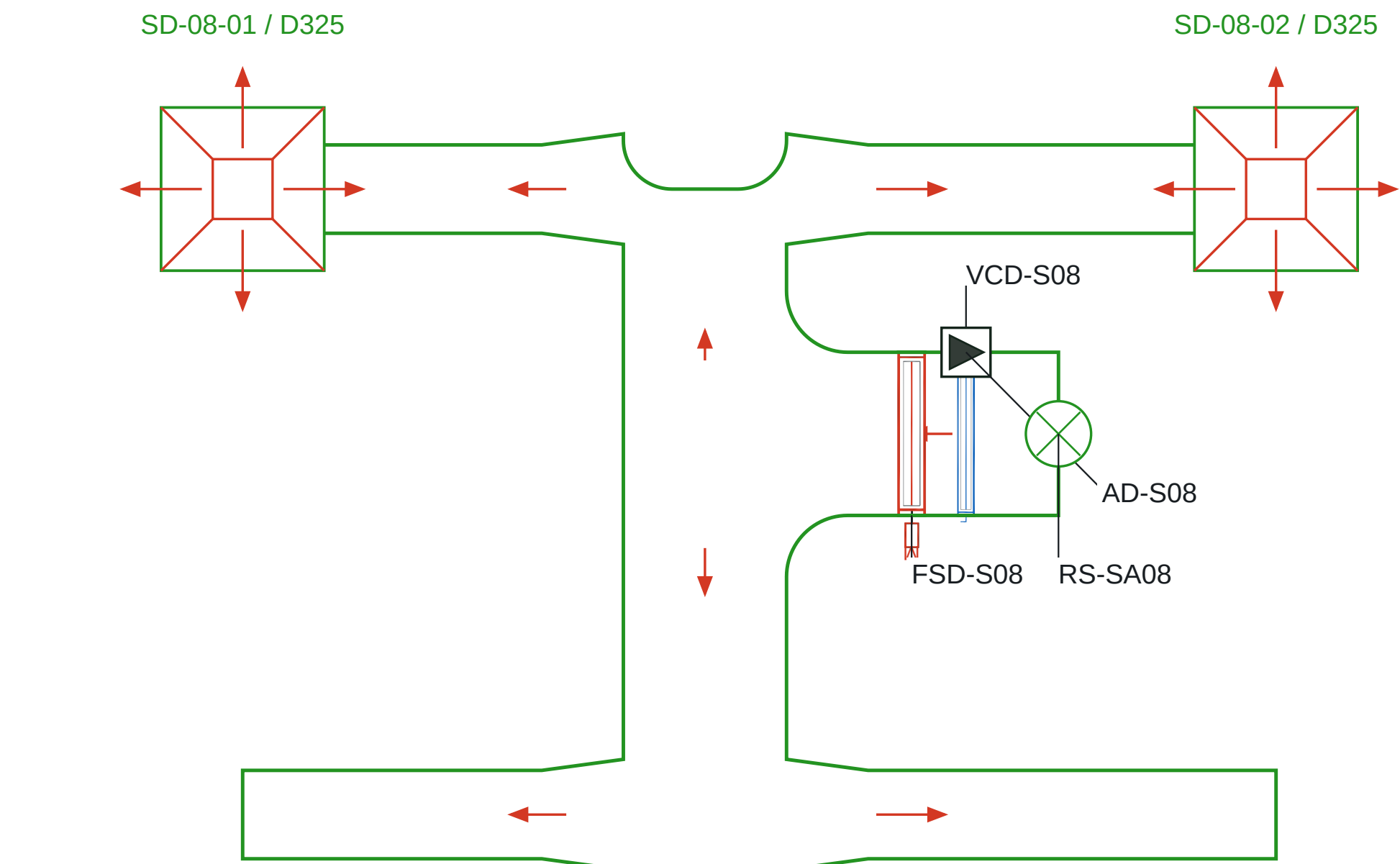
C MEETING CONTINUING HEADER / SIDE NECK / 1:20



Actual Rev11 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct\_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

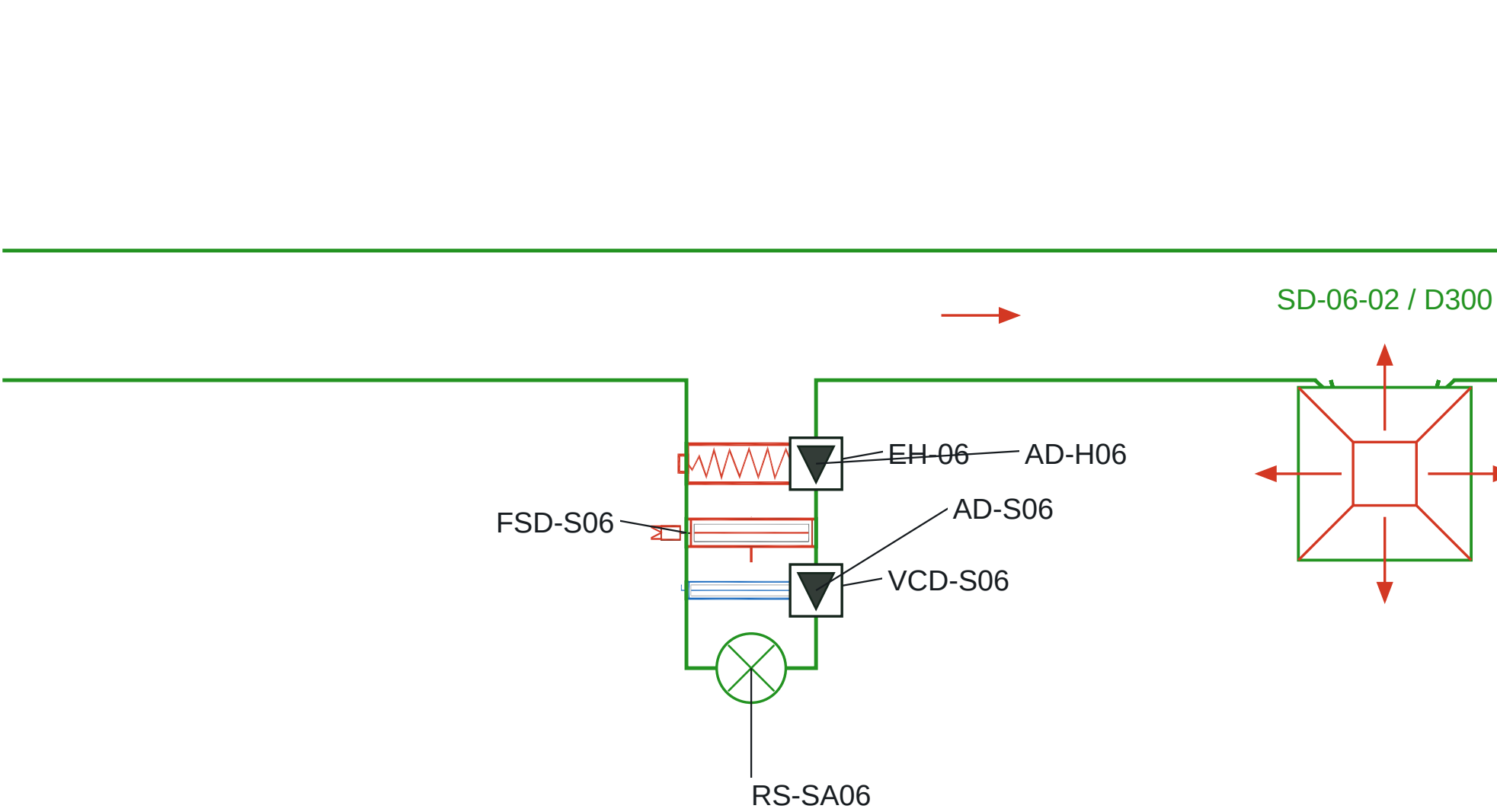
D TELECOM FEEDER / UP AND DOWN HEADERS / 1:20



Actual Rev11 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct\_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

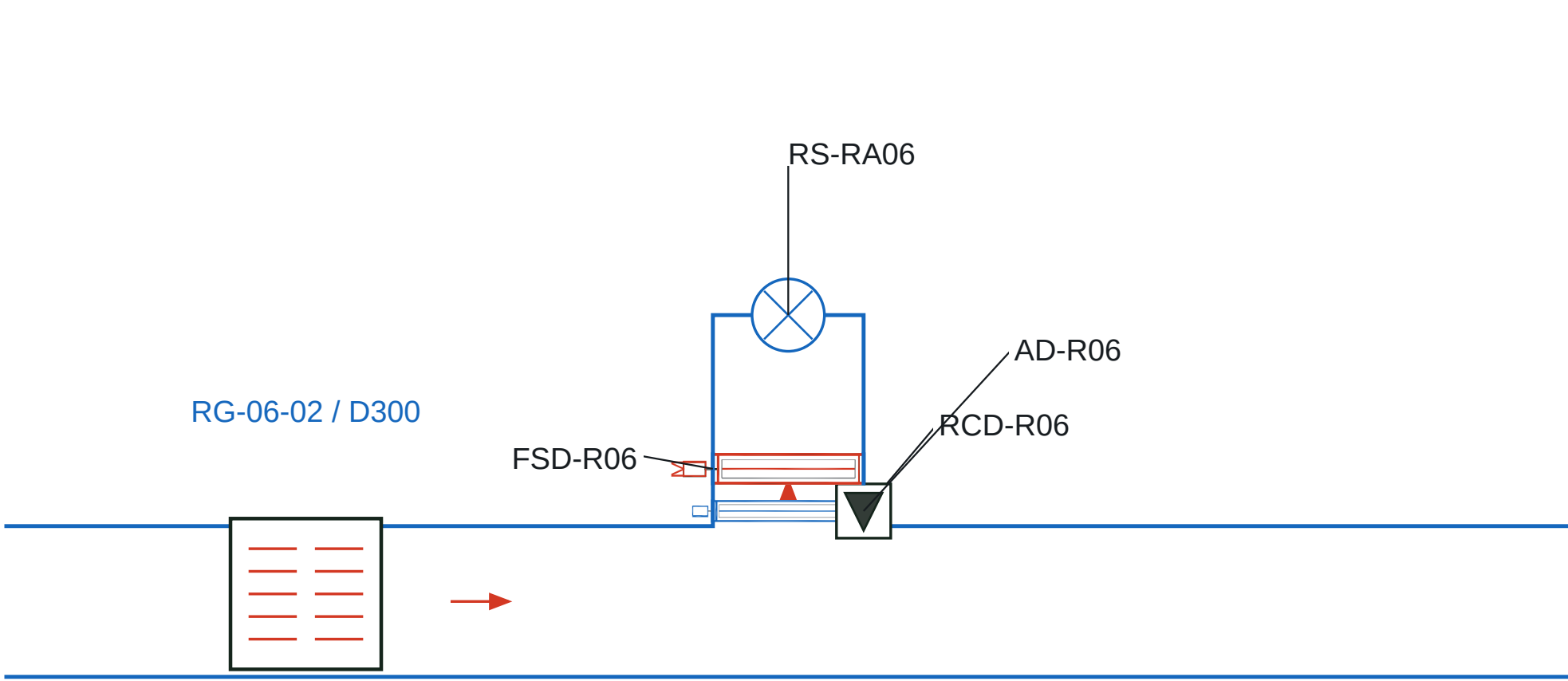
E CORRIDOR ROOM DROP BRANCH / HEATER / HEADER / 1:20



Actual Rev11 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct\_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

F CORRIDOR COLLECTION / RETURN DEVICE STATION / 1:20



Actual Rev11 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct\_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ACTUAL DUCT JUNCTIONS - ENLARGED PLAN DETAILS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-019

SCALE: AS SHOWN AT A1

REVISION

02

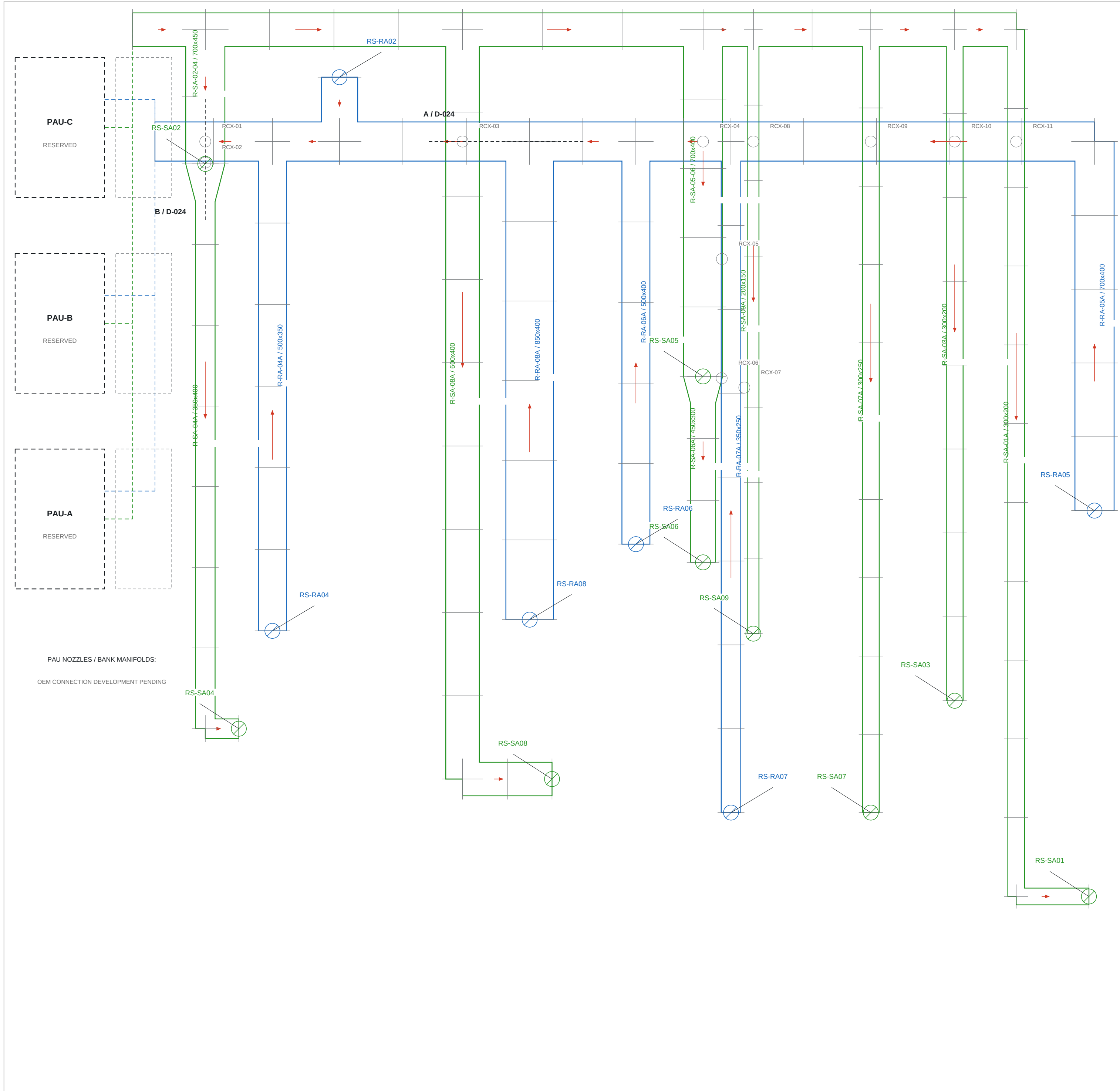
08 OCT 2026

SHEET 19 OF 50



## MAIN DUCT AND RETURN - ROOF DUCT ROUTING PROPOSAL

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033..039



## ROOF ROUTES / LEVELS

31 connected route segments feed all 15 room supply / return drops. Coordinates use the supplied floor plan origin. Roof footprint and PAU reservation zones must be surveyed.

GREEN SUPPLY: bare BOD +4.800 m; main 600 x 1300 mm. BLUE RETURN: bare BOD +5.550 m; main 700 x 900 mm. Tall roof mains limit plan width; wind / parapet screening and support stability require structural design.

75 mm outdoor insulation each face assumed. Eleven roof plan overlaps have separated insulation envelopes; minimum calculated separation 150 mm. No upper return riser intersects lower supply insulation in this proposed geometry.

PAU-A / B / C reservations: 1600 x 2500 mm each, 1000 mm east service strip. Dashed nozzle stubs and bank manifolds are schematic interfaces awaiting manufacturer dimensions

RCX tags: Roof\_crossings\_Rev12 register. A and B sections: D-024. Curbs and supports: D-030. Native centerline setting-out dimensions: D-032

| Service | Main clear mm | Bare BOD | Insulated top |
|---------|---------------|----------|---------------|
| SA      | 600 x 1300    | +4.800   | +6.175        |
| RA      | 700 x 900     | +5.550   | +6.525        |

ACCESS / LOAD HOLD

Service strips are reserved for PAUs only. Walking platforms, safe roof access, guarding, filter pull-out, common heater / conditional humidifier weather enclosure, banks and lifting path require selected equipment layout. Roof proposal is not an approved installation.

Roof levels use provisional roof surface +4.200 m; user underside +4.000 m plus assumed 200 mm slab. Route lengths are measured proposals, not surveyed as-built lengths. All sizes are clear internal mm.

## TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

## ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

## ROOF DUCT ROUTING PROPOSAL

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-023

SCALE: 1:50 AT A1

REVISION

02

08 OCT 2026

SHEET 23 OF 50



MAIN DUCT AND RETURN - ROOF CROSSING AND ROOM DROP SECTIONS

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033.039

A RCX-03 / SA08 x RA MAIN | 1:20

SA 600 x 400 / RA 700 x 900

RA BOD +5.550  
200 BETWEEN INSULATION AT RCX-03

SA BOD +4.800

ROOF +4.200\*

SLAB U/S +4.000

MINIMUM CEILING +3.300

\* Proposed dimensions / levels. Structural slab thickness, beam downstands, curb opening, firestop, weatherproof flashing, pipe / cable routes and support loads require coordination. No roof penetration may be set out from this proposal alone.

B RS-SA02 DROP | VERTICAL 1:20 / HORIZONTAL NTS

TWO ELBOWS / TRANSITION AT ROOF

BARE 500 x 350; RISE 1,400 m\*

ROOF BOD +4.800\*

CURB 300\* / WEATHER SEAL

ROOF +4.200\*

SLAB U/S +4.000

INDOOR BOD +3.400

MINIMUM CEILING +3.300

\* Proposed dimensions / levels. Structural slab thickness, beam downstands, curb opening, firestop, weatherproof flashing, pipe / cable routes and support loads require coordination. No roof penetration may be set out from this proposal alone.

VERTICAL DEVELOPMENT / CONNECTION RULES

Riser straight-length schedules use proposed bare BOD differences at equal-size horizontal ends. Elbow development and differing roof header sizes are separate fitting allowances. Maintain 3.300 m finished ceiling; do not lower a room duct to clear a roof route.

Use shop-developed mitred elbows with turning vanes or selected radiused elbows. Square plan junctions on the roof sheet denote fitted tees; they are not four-way crosses. Every room drop has an independent riser ID and reserved curb opening.

Sections are coordination details, 1:20 at A1 where stated; equipment and curb symbols are NTS. Clear opening schedules: Roof\_curbts\_Rev12. Structural capacity, waterproofing and fire ratings have not been approved.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ROOF CROSSING AND ROOM DROP SECTIONS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-024

SCALE: AS SHOWN AT A1

REVISION

02

08 OCT 2026

SHEET 24 OF 50



MAIN DUCT AND RETURN - FITTING DEVELOPMENT AND FABRICATION RULES

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033\_039

01 MITRED ELBOW / VANES

W x H: register

VANES: SHOP DESIGN

SQUARE THROAT / 45 DEG MITRES

02 THREE-LEG TEE

THREE LEGS ONLY, OPEN BRANCH THROAT

03 NECK / RISER TRANSITION

L >= delta / (2 tan 15 deg)

MAX 30 DEG INCLUDED ANGLE PROPOSED

ALL FITTINGS / DIMENSION SCHEDULE

Fitting\_design\_Rev12 contains 236 tagged loss / geometry allowances covering indoor path bends, joints, terminal adapters, balancing / conditional fire dampers and roof / riser connections. It is a design development register, not an OEM bill of materials.

For an actual two-dimensional reducer, proposed L = max(300 mm, max(abs(W1-W2), abs(H1-H2)) / (2 tan 15 deg)). Hold the bottom level only if the selected terminal / device height permits; otherwise show the offset in the shop section.

Roof main sizes are constant along each service spine, so no unshown step reductions occur. Branch width / height change at the tee is developed within a shop boot. Keep main and branch walls open at the connected throat.

Bend / tee lengths must fit the reserved straight station before moving any VCD, RCD, MFD, EH or access door. Reference CAD glyph dimensions are symbols; actual face-to-face lengths and actuator envelopes require an OEM selection.

| Fitting | Riser   | Clear W x H   | Elbows / K each  |
|---------|---------|---------------|------------------|
| F12-207 | RS-SA02 | 500.0 x 350.0 | 2 / 0.50 assumed |
| F12-209 | RS-RA02 | 650.0 x 400.0 | 2 / 0.50 assumed |
| F12-211 | RS-SA04 | 350.0 x 400.0 | 2 / 0.50 assumed |
| F12-213 | RS-RA04 | 500.0 x 350.0 | 2 / 0.50 assumed |
| F12-215 | RS-SA08 | 600.0 x 400.0 | 2 / 0.50 assumed |
| F12-217 | RS-RA08 | 850.0 x 400.0 | 2 / 0.50 assumed |
| F12-219 | RS-SA05 | 600.0 x 350.0 | 2 / 0.50 assumed |
| F12-221 | RS-RA05 | 700.0 x 400.0 | 2 / 0.50 assumed |
| F12-223 | RS-SA06 | 450.0 x 300.0 | 2 / 0.50 assumed |
| F12-225 | RS-RA06 | 500.0 x 400.0 | 2 / 0.50 assumed |
| F12-227 | RS-SA09 | 200.0 x 150.0 | 2 / 0.50 assumed |
| F12-229 | RS-SA07 | 300.0 x 250.0 | 2 / 0.50 assumed |
| F12-231 | RS-RA07 | 350.0 x 250.0 | 2 / 0.50 assumed |
| F12-233 | RS-SA03 | 300.0 x 200.0 | 2 / 0.50 assumed |
| F12-235 | RS-SA01 | 300.0 x 200.0 | 2 / 0.50 assumed |

ACTUAL ROOF BRANCH TAPERS - mm

| Route    | Inlet   | Outlet      | Developed L |
|----------|---------|-------------|-------------|
| R-SA-04A | 700x450 | 350.0x400.0 | 675         |
| R-SA-06A | 700x400 | 450.0x300.0 | 475         |

All K values on this issue are explicit preliminary allowances, not selected catalogue coefficients. Adapter\_details\_Rev12 retains the actual connected terminal adaptor coordinates / proposed lengths. Fabrication approval remains open.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

FITTING DEVELOPMENT AND FABRICATION RULES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-025

SCALE: NTS AT A1

REVISION

02

08 OCT 2026

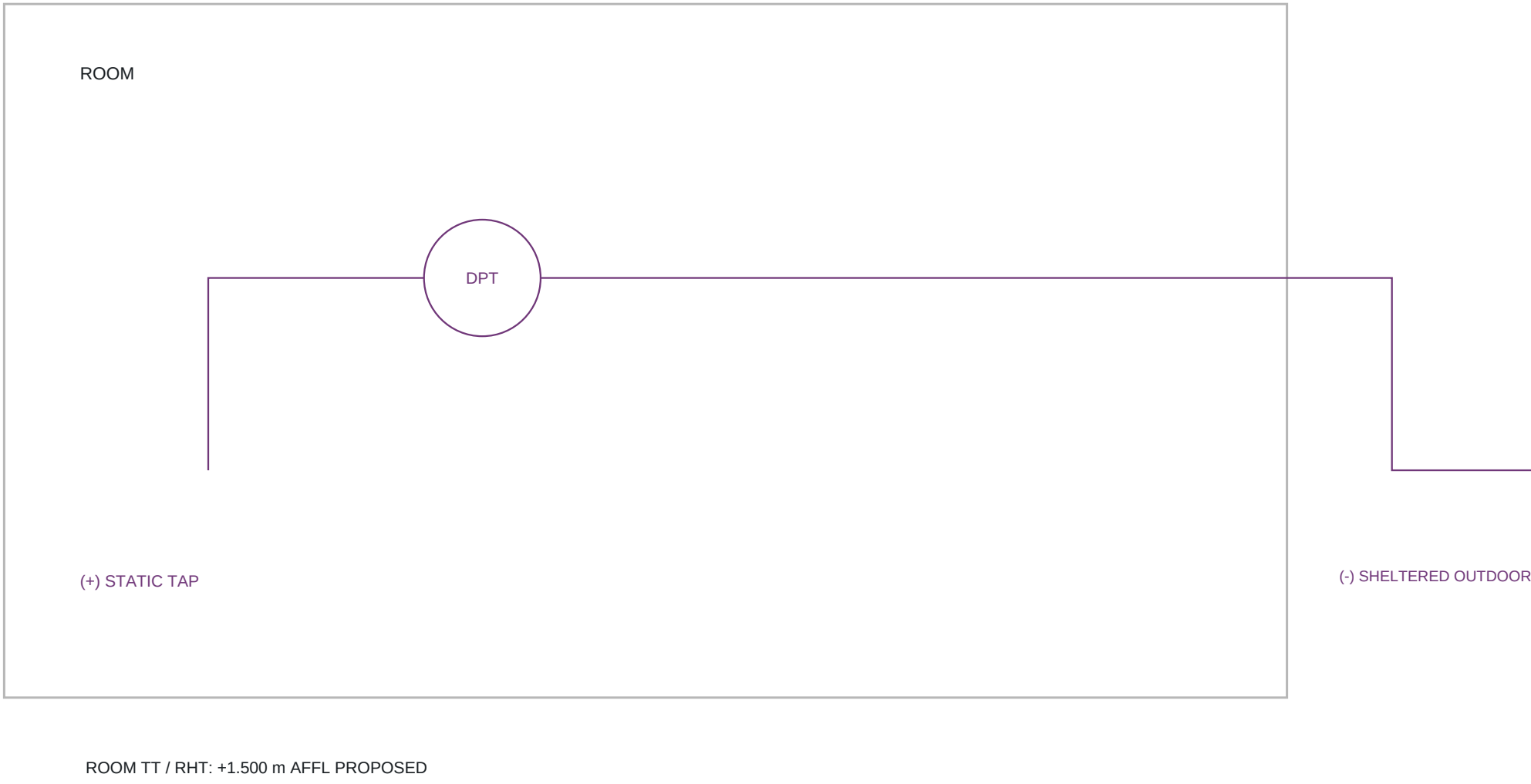
SHEET 25 OF 50



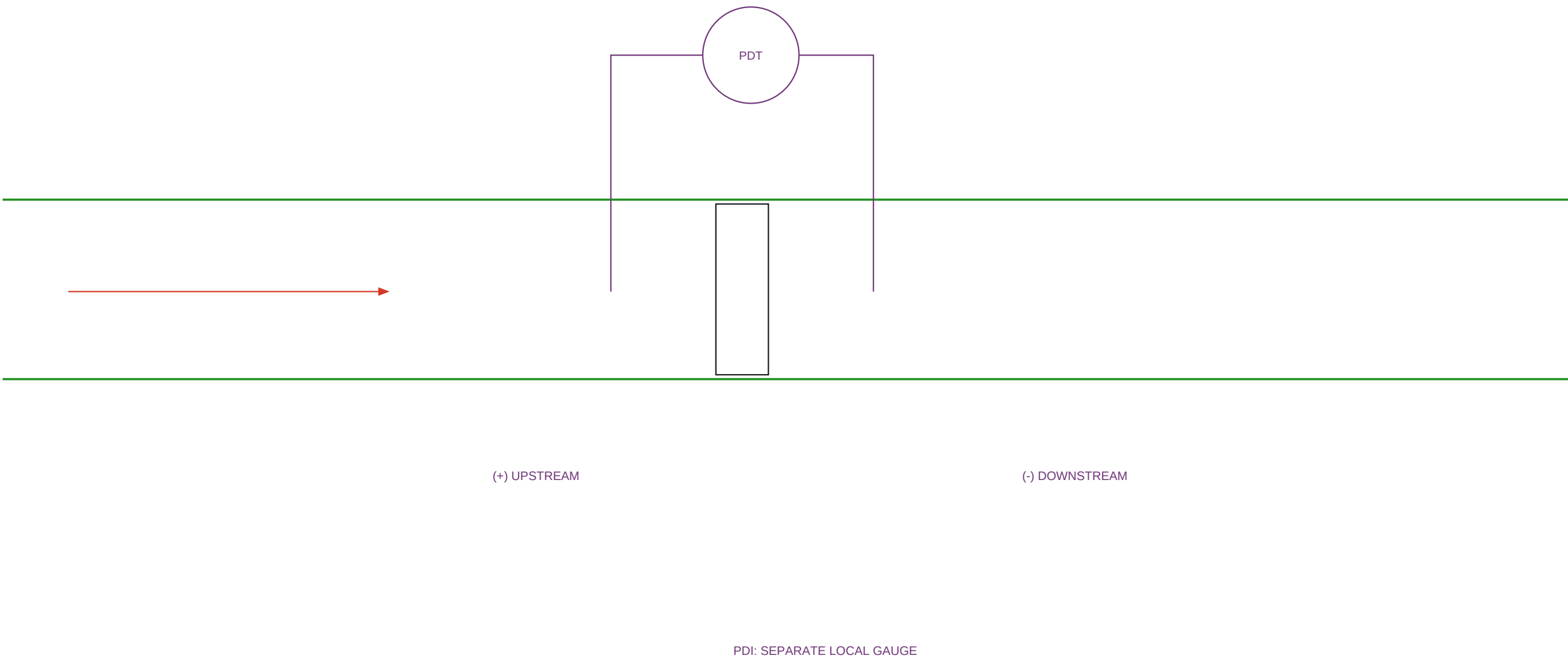
MAIN DUCT AND RETURN - INSTRUMENT MOUNTING AND PRESSURE TAP DETAILS

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033\_039

A ROOM STATIC PRESSURE



B FILTER / FAN DIFFERENTIAL



INSTALLATION SPECIFICATION - ALL 85 TAGS

Instrument\_installation\_Rev12 gives mounting, provisional ranges, pressure connections, wiring, access and commissioning requirements for every instrument. Tag requirement and I/O duties are retained from the approved input register, not replaced with generic symbols.

Proposed room DPT range -100..+100 Pa; fan / filter differential 0..1000 Pa; TT 0..50 degC; RH 0..100%; vibration 0..25 mm/s RMS. Ranges are selection proposals, not final alarm / trip settings.

Use labelled static pressure pickups and equal reference datums. Protect outdoor reference from wind / rain; avoid jet impingement, condensate traps and unserviceable tubing. Provide isolation and test ports per manufacturer.

COMMON-DUCT AIRFLOW / SENSOR ACCESS

FIT-SA-01 / FIT-RA-01: reserve 5Dh upstream and 3Dh downstream as a screening space allowance. Actual manufacturer measurement station / traversing method governs; no accuracy claim at the provisional roof position.

Main SA Dh = 0.821 m: screen 4.11 m upstream / 2.46 m downstream. Main RA Dh = 0.788 m: screen 3.94 m upstream / 2.36 m downstream. The current short bank-to-first-branch space does not prove these lengths; measurement station integration is an open hold.

Room temperature / humidity sensors avoid direct supply jets, exterior solar gain and heaters. VS installs on approved bearing / frame position. TSHH and AFS remain accessible for test / manual reset; selected heaters govern technology and settings.

Installation detail references D-028 / D-029 supplement D-012 to D-016 and the 85-row index. No final cable route, panel terminal number, approved sensor accuracy or OEM setpoint is asserted.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

INSTRUMENT MOUNTING AND PRESSURE TAP DETAILS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-028

SCALE: NTS AT A1

REVISION

02

08 OCT 2026

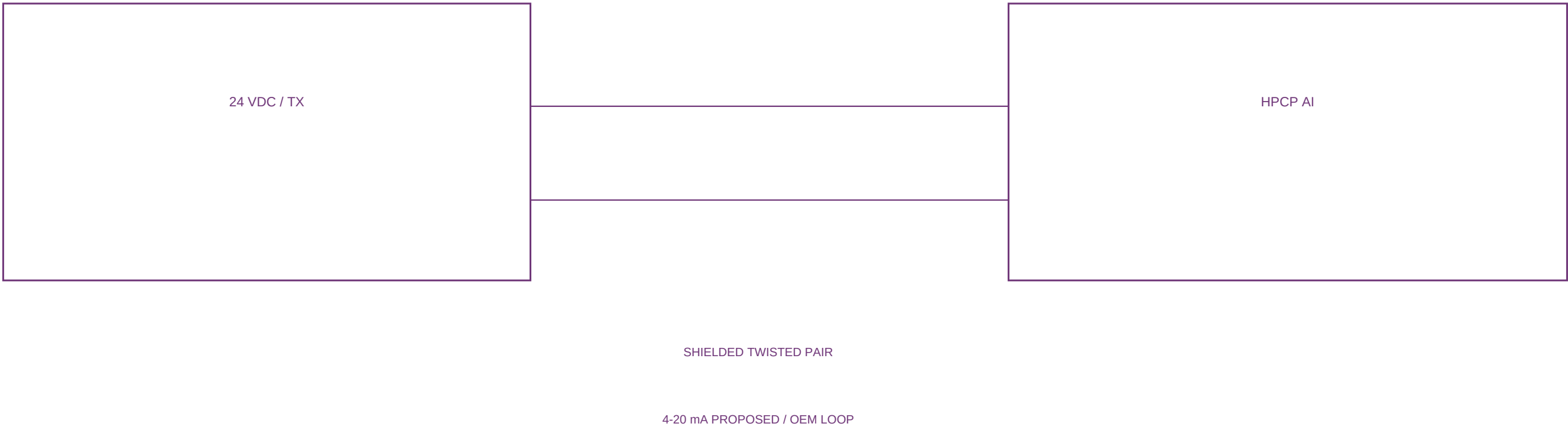
SHEET 28 OF 50



MAIN DUCT AND RETURN - I&C WIRING AND HEATER SAFETY INTERFACES

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033\_039

A ANALOG TRANSMITTER / AI



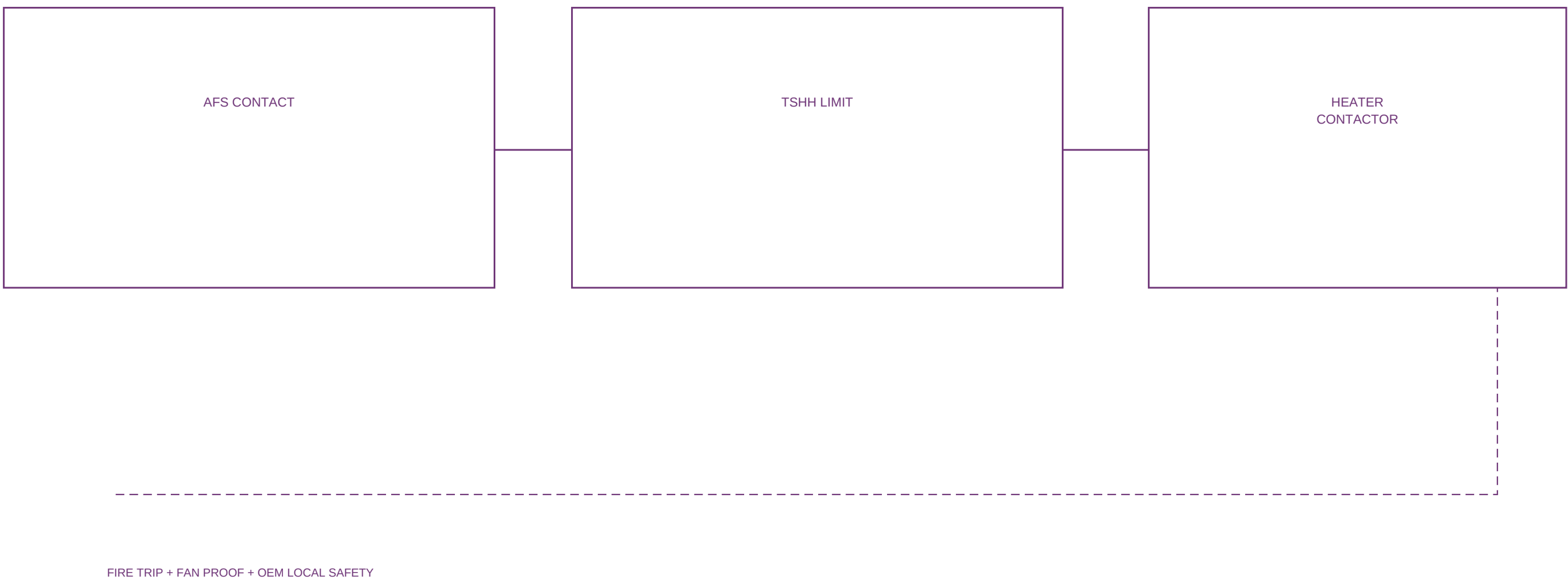
SIGNAL / PANEL REQUIREMENTS

AI proposals use isolated 4-20 mA channels and shielded twisted pairs. Confirm transmitter supply, two / three / four wire circuit, loop burden and earthing with OEM. Earth the cable shield at the panel end per project EMC design; segregate from mains / VFD cables.

DI proposals are dry contacts. Contact state, wetting voltage, galvanic isolation and fail-state indication require manufacturer wiring. Local PDI is not an AI. COMM energy meter protocol, CT ratio, address and termination remain pending.

Every tag maps to the unchanged 275-row I/O register. Final channel numbers, junction box locations and cable lengths await panel design and coordinated routes. Commission with traceable point-to-point checks and calibrated simulated signals.

B HARDWIRED HEATER PERMISSIVES



SAFETY / CAUSE AND EFFECT

Six AFS and six independent TSHH heater provisions remain present. Loss of airflow or high-limit trip inhibits heat through the OEM hardwired circuit. PLC heating demand cannot bypass either permissive. Alarm feedback reports to HPCP.

Confirmed fire stops all HVAC fans and heaters. No automatic fire extraction / purge is added. Fire damper rating, fail position, reset procedure and boundary quantity follow approved fire strategy. Manual reset is required after confirmed fire.

This is an interface diagram, not an electrical terminal drawing. Complete the heater OEM wiring, upstream electrical protection, safety relay architecture and fire interface review before procurement.

Input requirement and I/O assignments are retained. Final wiring and approved safety settings require selected hardware. Instrument\_Installation\_Rev12 is the 85-row specification companion.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

I&C WIRING AND HEATER SAFETY INTERFACES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-029

SCALE: NTS AT A1

REVISION

02

08 OCT 2026

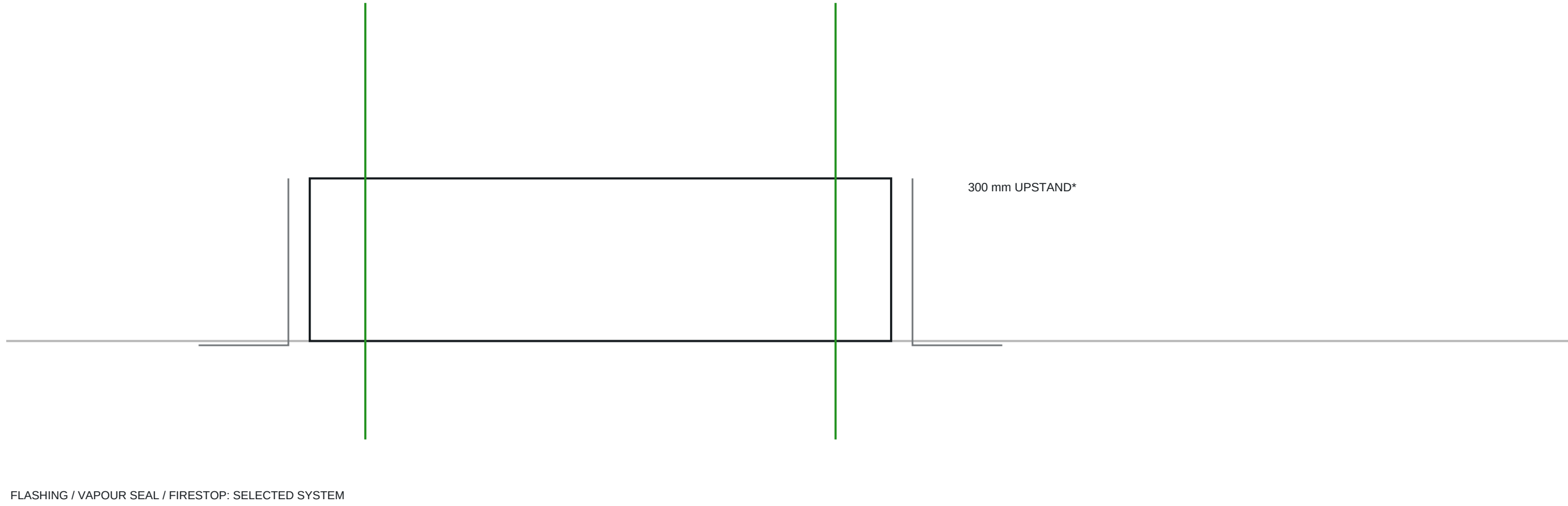
SHEET 29 OF 50



MAIN DUCT AND RETURN - ROOF CURB, SUPPORT AND WEATHERPROOFING

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033\_039

A PROPOSED CURB SECTION - NTS



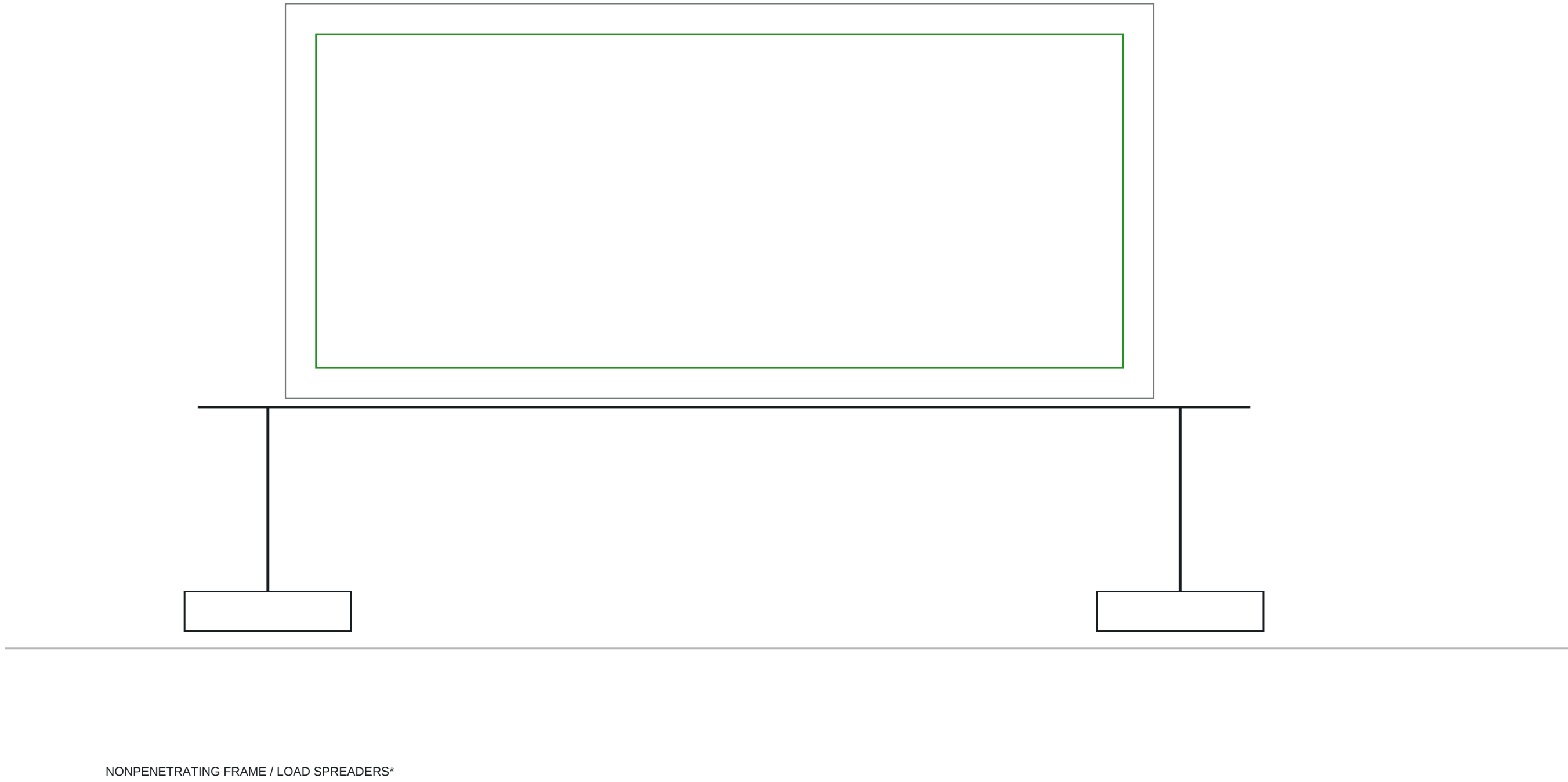
CURBS / PENETRATIONS

15 curb reservations have X / Y coordinates and opening envelopes: clear duct plus 75 mm insulation and 100 mm installation allowance each side. Upstand 300 mm above proposed roof surface; confirm finished waterproofing datum and local exposure.

Coordinate opening clearances, sleeves, firestop and insulated riser separation with structural and fire disciplines. Keep weatherproofing continuous; selected flashing / vapour seal and cladding joints must shed rain. Do not cut a slab or beam from this drawing.

Outdoor insulation 75 mm is provisional. Specify thermal conductivity, vapour barrier, sealed cladding, weather / UV resistance and fire classification from approved project materials. It is not a declared UAE average.

B ADJUSTABLE SUPPORT FRAME - NTS



SUPPORT / ACCESS PROCUREMENT HOLD

157 support station reservations, maximum nominal 1500 mm route spacing. Deduplicate junction stations and adjust fitting / equipment supports in shop drawings. Geometry alone does not select a bracket or roof anchor.

Support load table remains TBC until gauge, fitting weight, insulation / cladding, actuator, wind, seismic requirements and frame self-weight are known. Tall roof mains need wind stability / bracing and maintenance platform design.

Verify structure and membrane bearing / protection before choosing nonpenetrating ballast or anchors. PAU lifting, coil / filter withdrawal and heater / humidifier service envelopes require a complete manufacturer footprint.

\* Proposal only, Roof\_curbs\_Rev12 and Roof\_supports\_Rev12 provide setting-out reservations, not anchor design or certified structural loads. Roof surface +4.200 m assumes 200 mm structural slab above the confirmed +4.000 m underside.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ROOF CURB, SUPPORT AND WEATHERPROOFING

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-030

SCALE: NTS AT A1

REVISION

02

08 OCT 2026

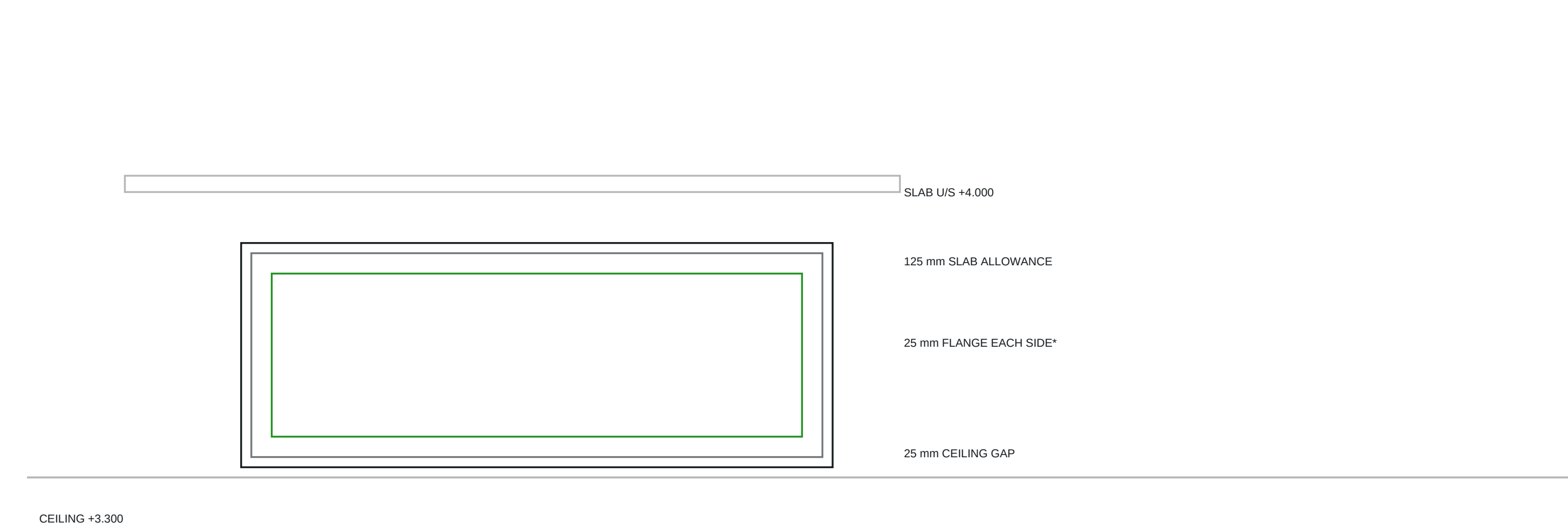
SHEET 30 OF 50



MAIN DUCT AND RETURN - CEILING COMPONENT AND ACCESS ENVELOPES

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033\_039

A DEEPEST DUCT / FLANGE | 1:10



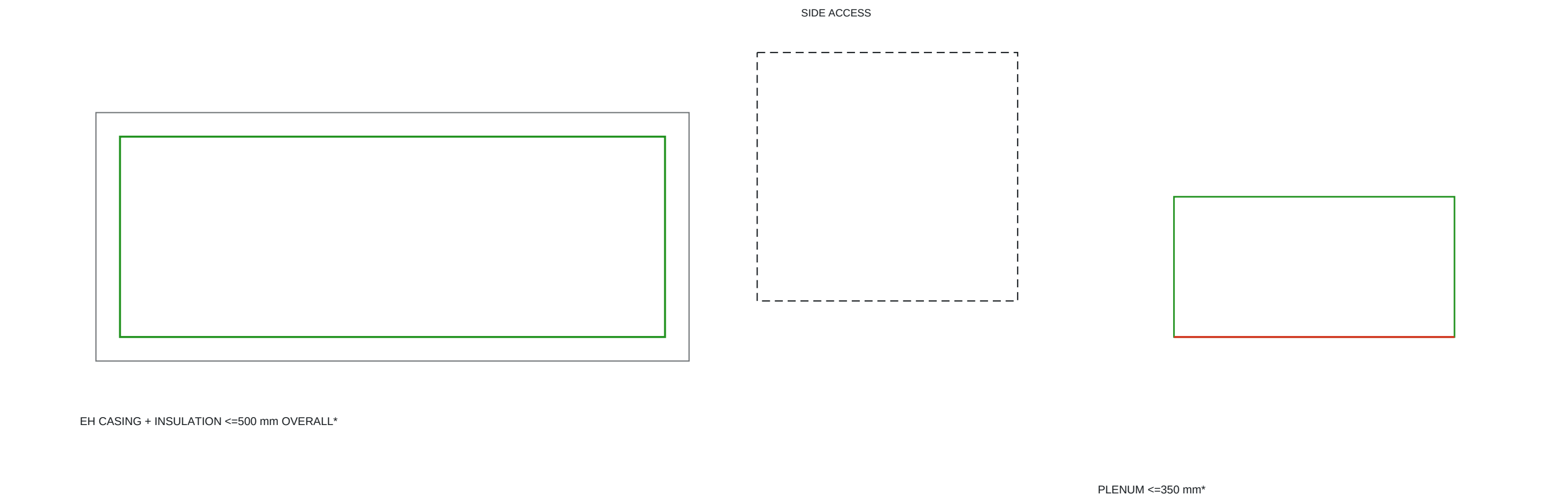
DUCT + FLANGES

Deepest bare depth 400 mm, BOD +3.400 m, 50 mm insulation each face, proposed 25 mm flange allowance beyond insulation: overall reserved depth 550 mm. Lowest +3.325 m; highest +3.875 m. Ceiling gap reduces to 25 mm; slab allowance reduces to 125 mm.

This is a fit screen, not a maintenance clearance. Hanger / beam / other-service and flange shape / vapour seal require shop coordination. An actuator above the duct can exceed this envelope: select a side actuator / lower-profile assembly or locally reroute.

Installation\_envelopes\_Rev12 checks every indoor duct and specifies room-heater procurement limits. No selected component envelope is represented as verified.

B ROOM HEATER / PLENUM | NTS



HEATER / TERMINAL PROCUREMENT LIMITS

Room heater casing plus insulation must fit between +3.350 and +3.850 m (500 mm overall) unless separately coordinated. Reserve side service access and an aligned ceiling hatch for reset / removal. OEM isolation and thermal clearances govern.

A 350 mm overall terminal plenum above ceiling +3.300 m reaches +3.650 m. Provide an offset / vertical neck to the branch; verify the actual plenum, collar and flex radius in the 700 mm ceiling void. A 600 mm face is not a 600 mm-deep plenum.

All access doors and actuators need a removal path. Final hatch dimensions, heater face-to-face length and damper / actuator position cannot be approved from symbolic plan blocks. If OEM envelopes exceed limits, revise the local branch before release.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

CEILING COMPONENT AND ACCESS ENVELOPES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-031

SCALE: AS SHOWN AT A1

REVISION

02

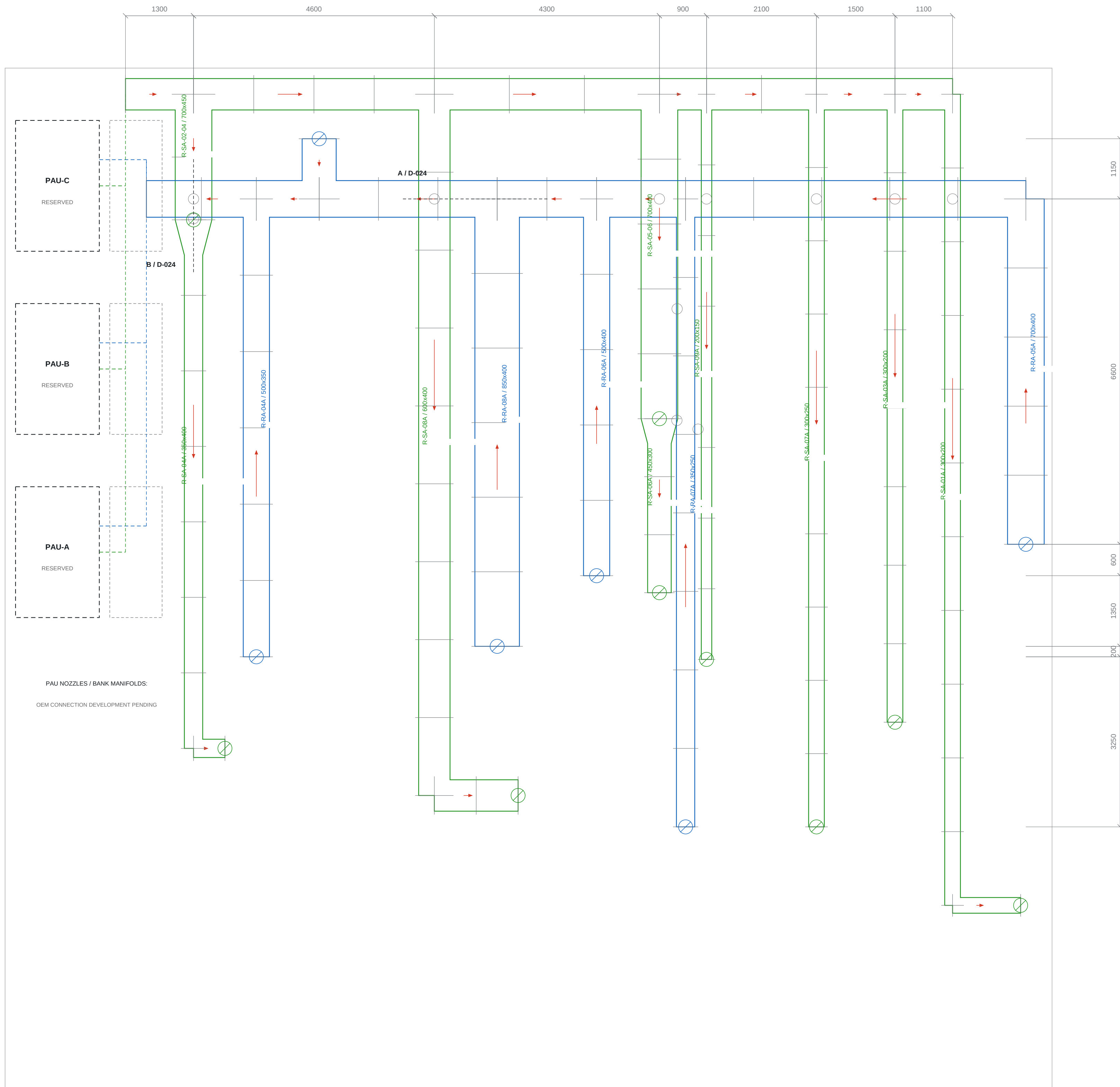
08 OCT 2026

SHEET 31 OF 50



### MAIN DUCT AND RETURN - ROOF SETTING-OUT AND ROOM DROP REGISTER

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033..035



**PROPOSED ROOF DROP COORDINATES - mm**

| Riser   | X     | Y     | Clear mm  | Rise m* |
|---------|-------|-------|-----------|---------|
| RS-SA02 | 3600  | 16600 | 500 x 350 | 1.4     |
| RS-RA02 | 6000  | 18150 | 650 x 400 | 2.15    |
| RS-SA04 | 4200  | 6500  | 350 x 400 | 1.4     |
| RS-RA04 | 4800  | 8250  | 500 x 350 | 2.15    |
| RS-SA08 | 9800  | 5600  | 600 x 400 | 1.4     |
| RS-RA08 | 9400  | 8450  | 850 x 400 | 2.15    |
| RS-SA05 | 12500 | 12800 | 600 x 350 | 1.4     |
| RS-RA05 | 19500 | 10400 | 700 x 400 | 2.15    |
| RS-SA06 | 12500 | 9475  | 450 x 300 | 1.4     |
| RS-RA06 | 11300 | 9800  | 500 x 400 | 2.15    |
| RS-SA09 | 13400 | 8200  | 200 x 150 | 1.4     |
| RS-SA07 | 15500 | 5000  | 300 x 250 | 1.4     |
| RS-RA07 | 13000 | 5000  | 350 x 250 | 2.15    |
| RS-SA03 | 17000 | 7000  | 300 x 200 | 1.4     |
| RS-SA01 | 19400 | 3500  | 300 x 200 | 1.4     |

## SETTING-OUT / REGISTER LINKS

13 native roof dimension segments supplement the 11 retained architectural dimensions. Individual route end coordinates and lengths are in Roof\_routes\_Rev12; curbs use the same XY origin.

Local origin is the supplied floor drawing (0,0), not a surveyed benchmark. Verify parapets, equipment plinths, openings, wind screening, beam grids and vertical datums before establishing roof setting-out.

\* Rises use provisional roof levels; bend / reducer development and selected PAU nozzle connections are separate. Bank interface stubs are schematic, not dimensioned equipment outlets.

Source footprint 20000 x 19500 mm. Dimensions indicate proposed route centerline coordinates. Nominal units mm / m AFFL; proposed heights and fabrication details require professional review and coordinated OEM submissions.

## TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

## ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

## ROOF SETTING-OUT AND ROOM DROP REGISTER

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-032

SCALE: 1:50 AT A1

REVISION

02

08 OCT 2026

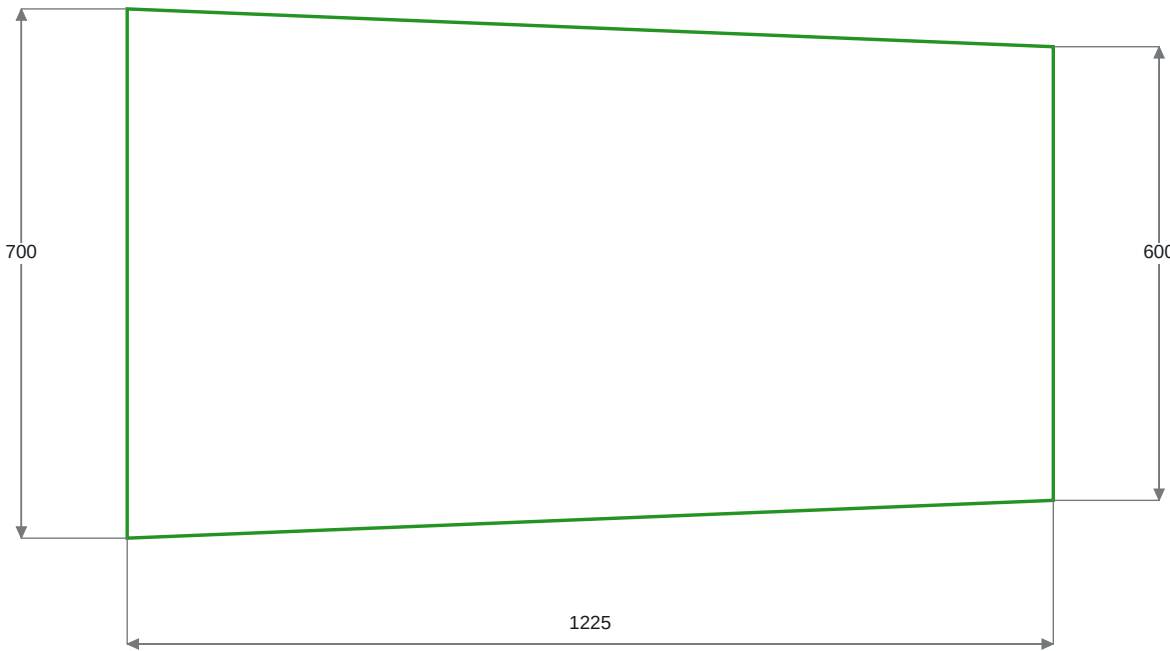
SHEET 32 OF 50



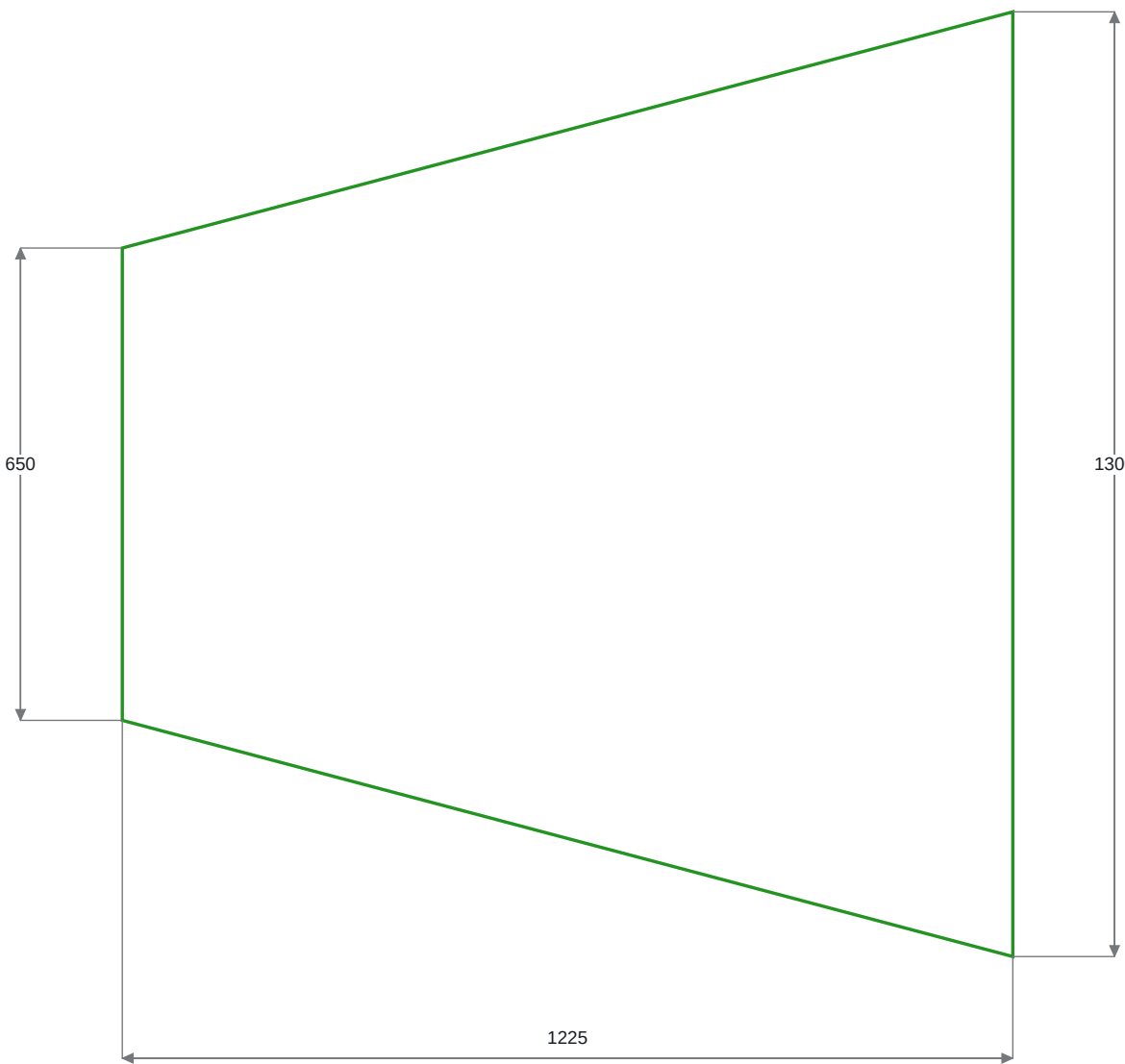
MAIN DUCT AND RETURN - DX PAU BRANCH / HEADER TRANSITION DEVELOPMENT

SA / PER DUTY UNIT / COORDINATION PROPOSAL

PLAN - CLEAR WIDTHS



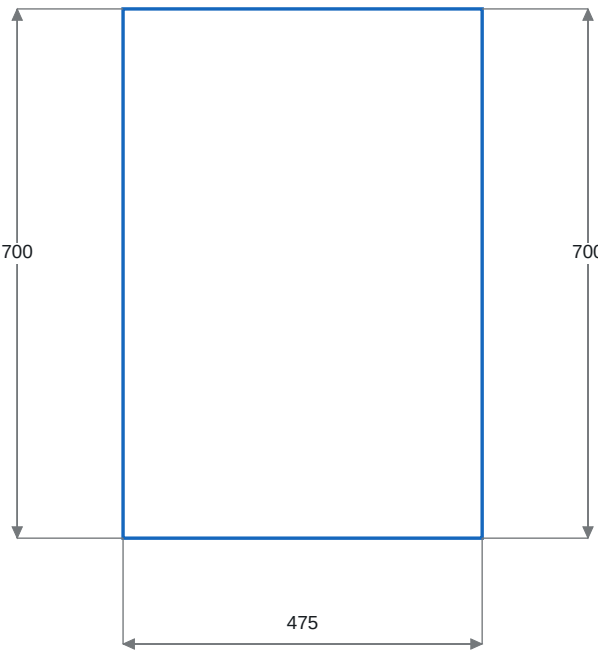
ELEVATION - CLEAR DEPTHS



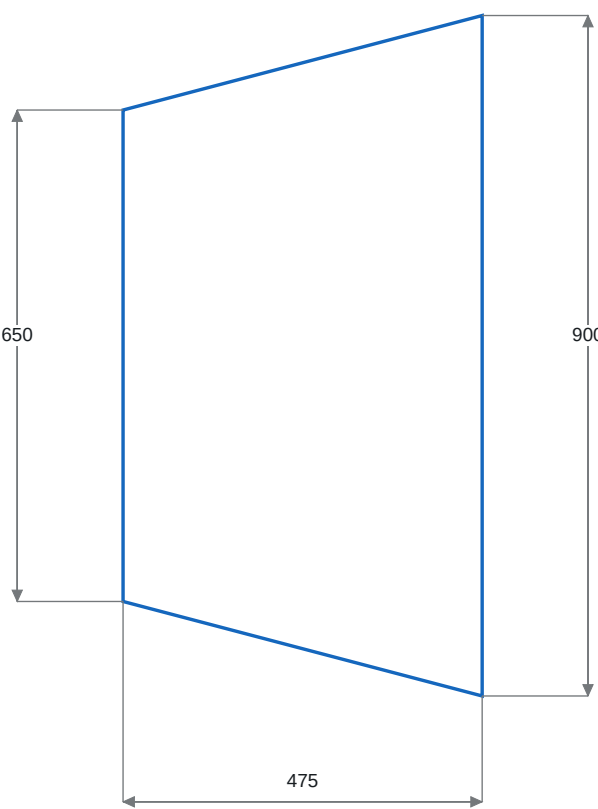
NOZZLE / FLEX / DAMPER / TEE BODY: OEM DEVELOPMENT

RA / PER DUTY UNIT / COORDINATION PROPOSAL

PLAN - CLEAR WIDTHS



ELEVATION - CLEAR DEPTHS



NOZZLE / FLEX / DAMPER / TEE BODY: OEM DEVELOPMENT

| Connection | Duty L/s | Branch W x H | Header W x H | m/s   | Taper L* |
|------------|----------|--------------|--------------|-------|----------|
| DX-SA-A    | 1848.90  | 700 x 650    | 600 x 1300   | 4.064 | 1225     |
| DX-RA-A    | 1503.34  | 700 x 650    | 700 x 900    | 3.304 | 475      |
| DX-SA-B    | 1848.90  | 700 x 650    | 600 x 1300   | 4.064 | 1225     |
| DX-RA-B    | 1503.34  | 700 x 650    | 700 x 900    | 3.304 | 475      |
| DX-SA-C    | 1848.90  | 700 x 650    | 600 x 1300   | 4.064 | 1225     |
| DX-RA-C    | 1503.34  | 700 x 650    | 700 x 900    | 3.304 | 475      |

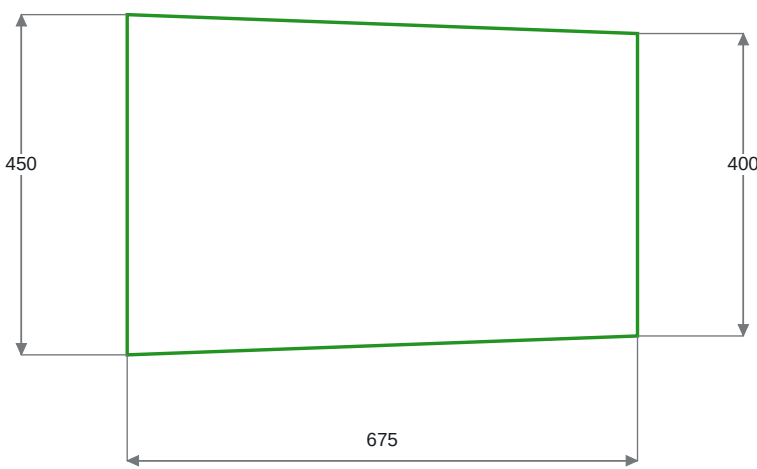
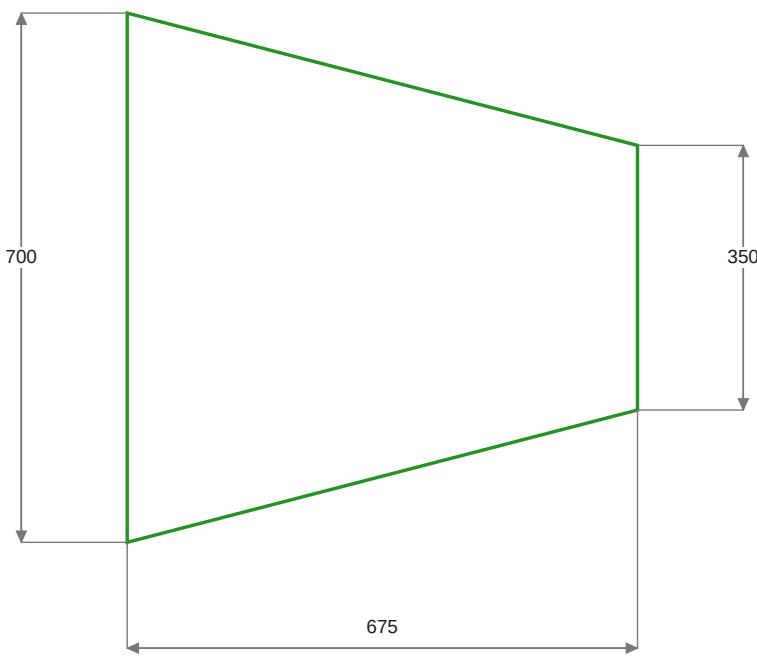
\* Concentric development L >= max(300, max(delta W,delta H)/(2tan15deg)), rounded up to25 mm. Dimensions are proposed spool development, not selected PAU ports or fabrication release. Three branches, maximum two duty units. Header/branch joining and pressure losses remain OEM/shop coordination.



MAIN DUCT AND RETURN - ACTUAL ROOF SPLIT / REDUCER / ELBOW DEVELOPMENT

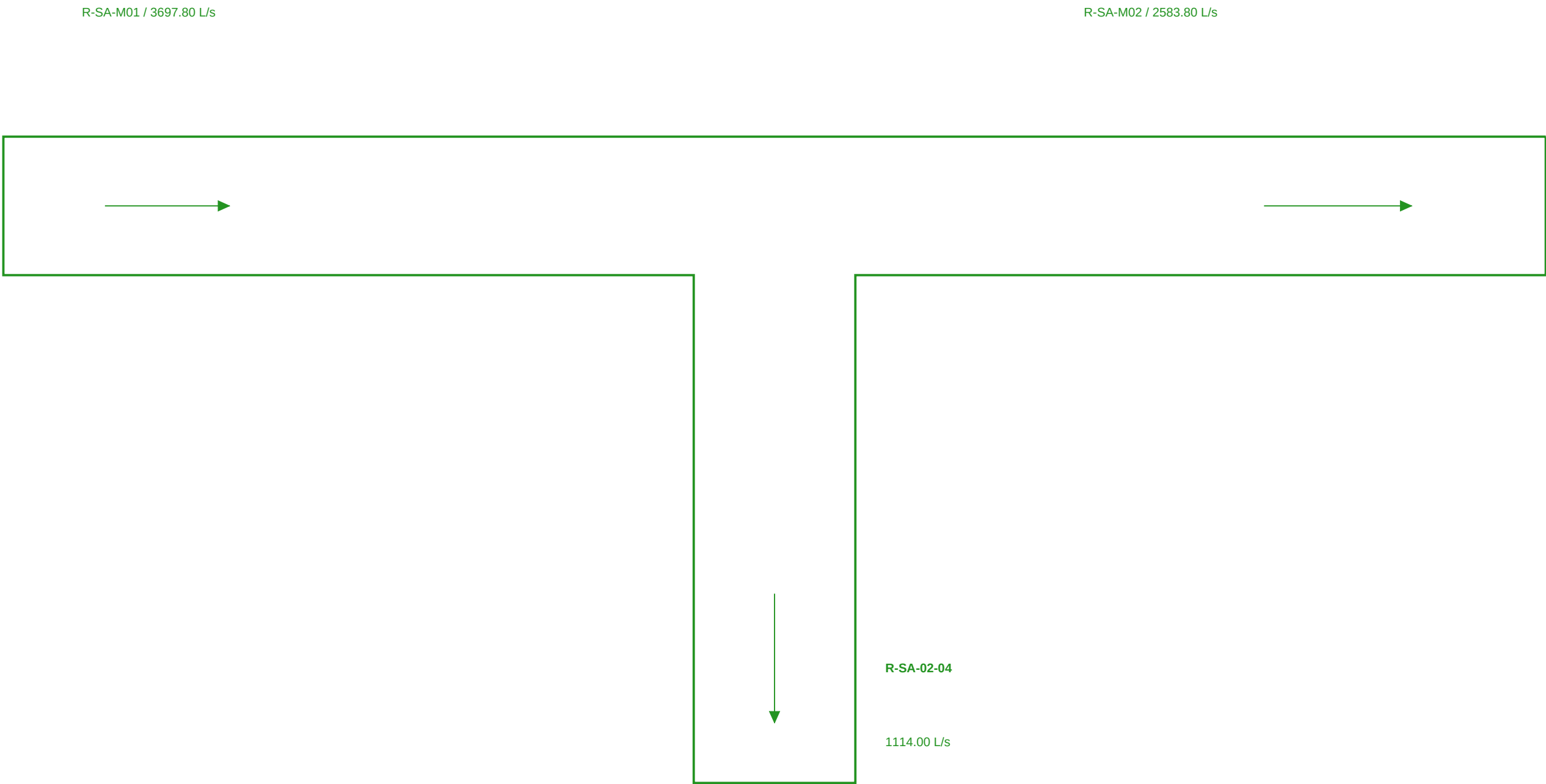
A R-SA-04A: ACTUAL REGISTERED TAPER

700 x 450 -> 350 x 400 / L675 / 449.20 L/s



PLAN ABOVE / ELEVATION BELOW / CLEAR INTERNAL mm

B ROOF SPLIT AT X3600 / Y19000



Open connected throat shown. SA spine600x1300; split branch700x450. Plan boot/radius and depth change are shop proposals. Keep bare BOD+4.800 m; coordinate every insulated offset with MDR-023/024 before release.

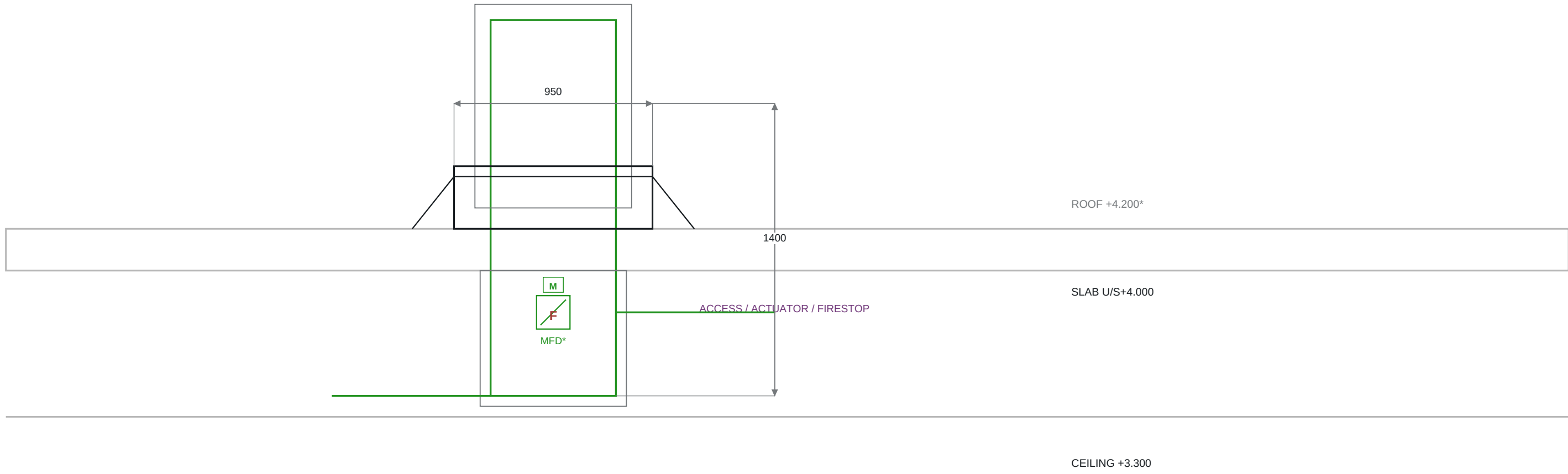
ELBOW / TRANSITION / ACCESS DEVELOPMENT

| Item            | Engineering requirement                                                       | Coordination evidence                                         |
|-----------------|-------------------------------------------------------------------------------|---------------------------------------------------------------|
| Reducer         | Concentric half-angle <=15deg proposed; actual space and fitted K checked     | R-SA-04A taper675 retained; other route tapers in Roof_routes |
| Radius elbow    | Proposed centerline radius >= clear width; vanes per selected construction    | Actual bend quantities / radii in Fitting_design_Rev12        |
| Boot / tee      | Keep connected openings; no line across throat; detail all three elevations   | Original actual room junction enlargements MDR-019            |
| Damper / access | Device body + actuator + removal space; inspection hatch reachable            | Components / Device_locations / MDR-028,031                   |
| Construction    | Pressure class, gauge, reinforcement, seam and leakage class to approved spec | OEM/SMACNA-based shop design, project approval pending        |

A uses the retained physical route dimensions. B is a development diagram of the registered split, not a new roof route. Actual duct profiles, independent-service separation and riser locations remain as MDR-001/023.



RS-SA08 / TELECOM / 600x400



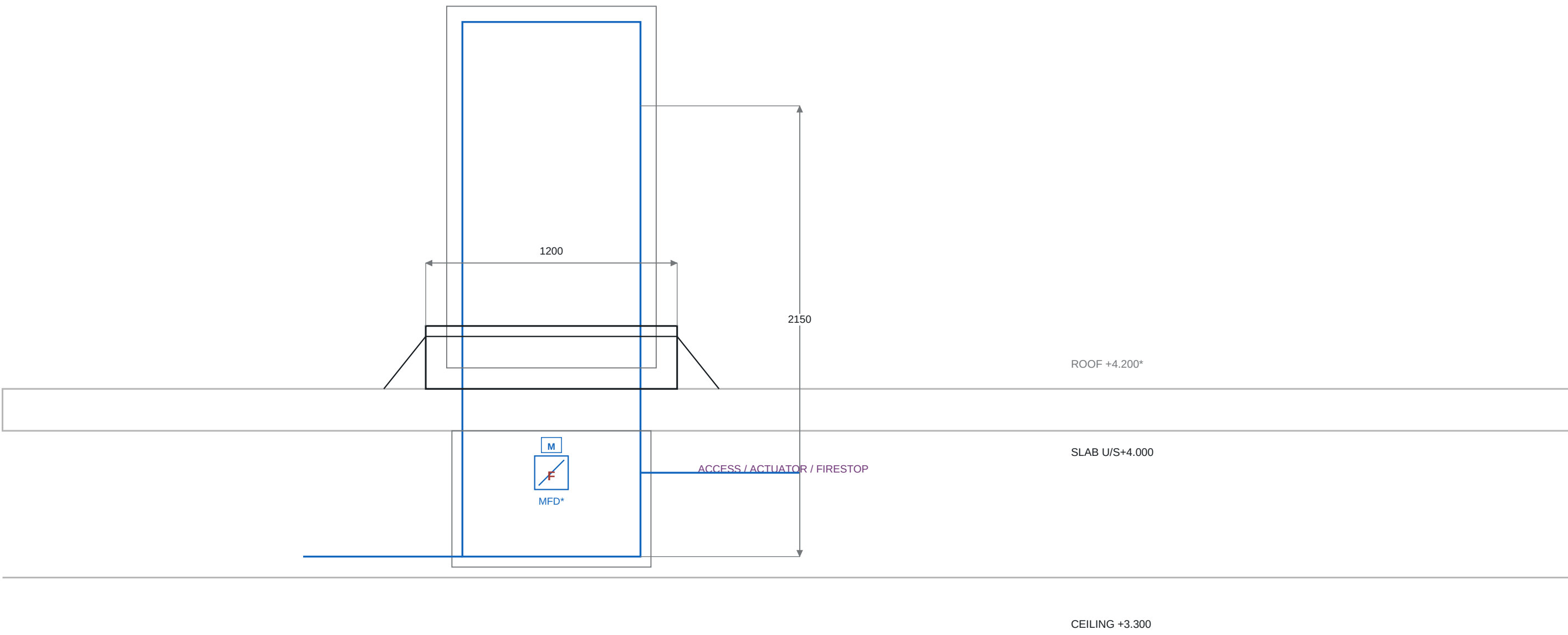
CURB OPENING 950.0 x 750.0 mm\*

Diagram locates the reserved shaft, curb300 mm upstand, weather flashing and conditional fire damper. Duct cross-section width / level rise are scaled; exact 3D elbow, vertical-transition/roof-to-room insulation joint, sleeve/firestop and selected damper body must be shop-developed.

For all15 shaft coordinates / openings use MDR-022/032 and the Roof\_risers / Roof\_curbs tables. Local depth400 with50 mm insulation +25 mm flange allowance leaves25 mm ceiling gap and125 mm slab allowance; actual components/beam access remain open.

\* Roof+4.200 assumes200 mm slab; outdoor75 / indoor50 mm insulation and curb/opening allowances are proposals. Installed elevations not assigned. RA05 curb extends25 mm beyond nominal footprint: resolve roof/parapet/shaft setting-out before approval.

RS-RA08 / TELECOM / 850x400



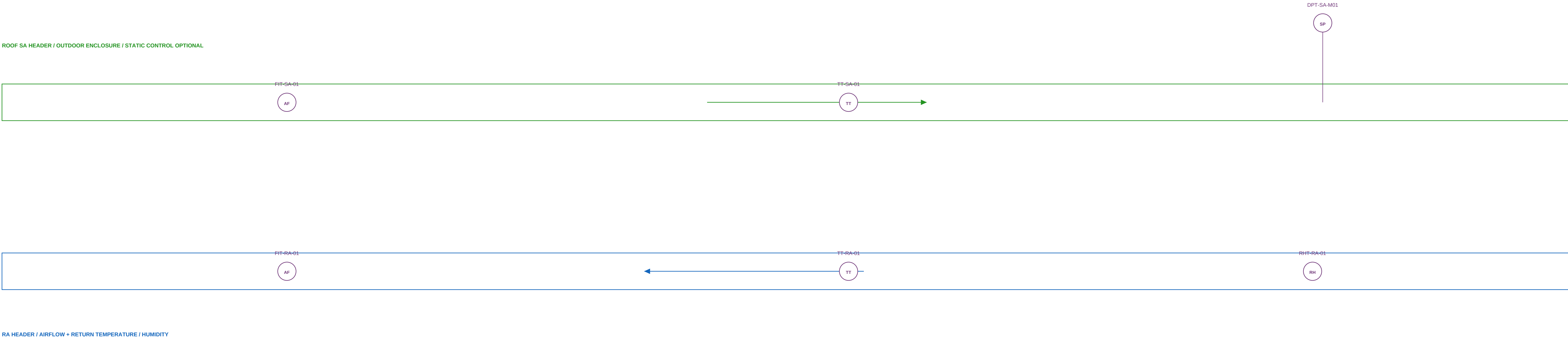
CURB OPENING 1200.0 x 750.0 mm\*

Diagram locates the reserved shaft, curb300 mm upstand, weather flashing and conditional fire damper. Duct cross-section width / level rise are scaled; exact 3D elbow, vertical-transition/roof-to-room insulation joint, sleeve/firestop and selected damper body must be shop-developed.

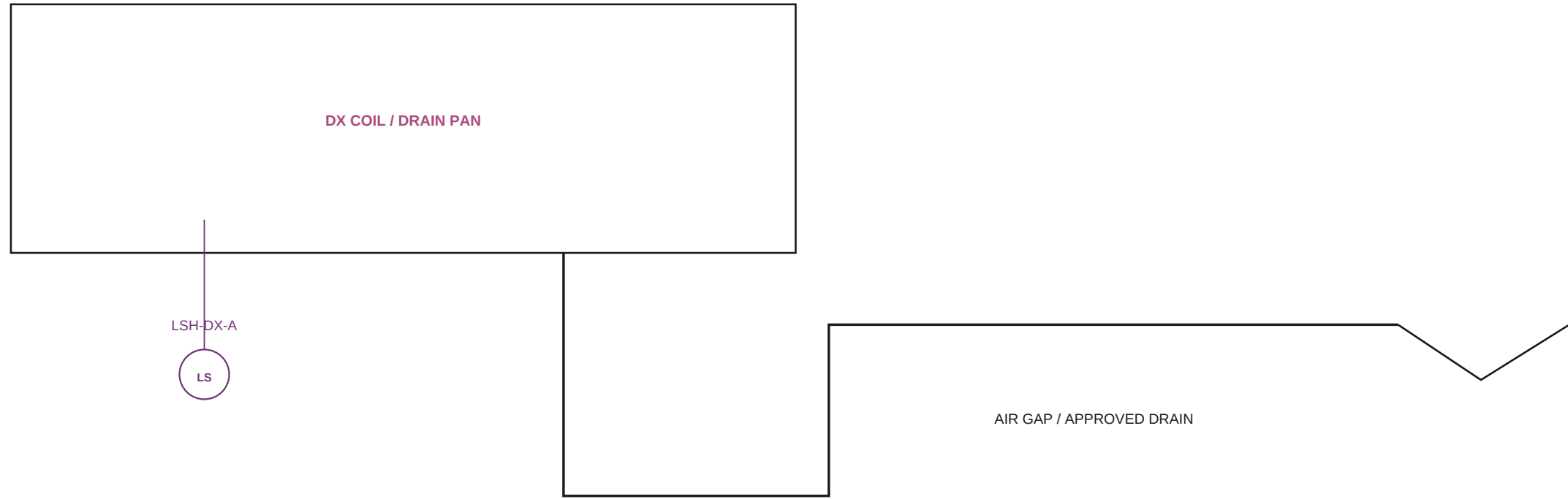
For all15 shaft coordinates / openings use MDR-022/032 and the Roof\_risers / Roof\_curbs tables. Local depth400 with50 mm insulation +25 mm flange allowance leaves25 mm ceiling gap and125 mm slab allowance; actual components/beam access remain open.



A COMMON AIR STATIONS AND SENSORS



B NEGATIVE-PRESSURE DRAIN-PAN TRAP / SERVICE



Trap seal / leg dimensions selected from the actual maximum negative pan pressure and OEM instructions; do not select from the provisional fan budget. Prime, provide cleaning access, slope continuous drain, prevent overflow and protect insulation/vapour seals. Apply separately to PAU-A/B/C.

C MOUNTING / MEASUREMENT / ACCESS RULES

| Point          | Detailed requirement                                                                                |
|----------------|-----------------------------------------------------------------------------------------------------|
| FIT            | Selected station straight runs / calibration and min-flow check; bank elbows can invalidate station |
| TT/RHT         | Serviceable insertion, representative air; no heater radiant bias / condensate contact              |
| Static DPT     | Flush static pickup, sealed labelled tubing, weather enclosure; station/setpoint OEM review         |
| Pressure taps  | DPT-SF across fan; filter PDT/PTDI tap pairs; identify H/L, isolation/testing access                |
| DX refrigerant | Pressure taps / transducers only at approved factory service points                                 |
| Maintenance    | Filter pull-out, coil face, trap, compressor, damper and electrical access to OEM drawings          |

AIR85 installation specifications are restored in the registers and MDR-028/029. This sheet adds the DX/common-header requirements. LSH example uses PAU-A tag; B/C have the same factory-reviewed function. No trap dimensions or refrigerant service taps are fabricated.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

COMMON HEADER INSTRUMENTS / CONDENSATE / ACCESS

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-036

SCALE: NTS AT A1

DRAWING REVISION 02

08 OCT 2026

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MAIN DUCT AND RETURN - DX / FAN / DAMPER / FIRE CAUSE AND EFFECT

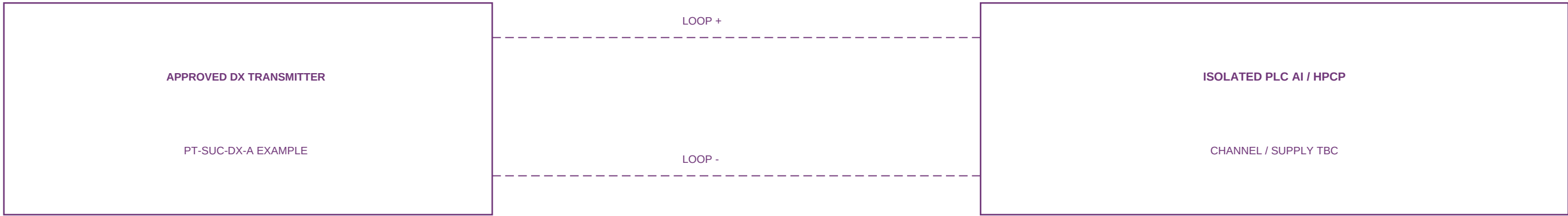
| Event / references                                                                                          | DX / heating                                                       | Fans / maximum duty                                          | Dampers / isolation                                         | Reset                                |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------|
| Confirmed fire / E-stop<br>FIRE/ESD                                                                         | Stop all affected DX equipment through approved fire/OEM interface | Stop ALL HVAC fans; heaters OFF                              | Approved fire positions; not inferred from normal isolation | Manual fire reset after clearance    |
| Compressor high pressure<br>PSH-DX-A / PSH-DX-B / PSH-DX-C                                                  | OEM hardwired cutout                                               | OEM safe response; failed unit/changeover as approved        | Isolate failed unit if stopped                              | OEM latch/reset threshold TBC        |
| Low pressure<br>PSL-DX-A / PSL-DX-B / PSL-DX-C                                                              | OEM operating-envelope protection                                  | OEM routine; no bypass                                       | OEM safe state                                              | OEM reset / delay TBC                |
| Evaporator frost<br>TSL-L-DX-A / TSL-L-DX-B / TSL-L-DX-C                                                    | Inhibit / stage through OEM                                        | OEM defrost/anti-freeze routine                              | Approved OEM positions                                      | OEM anti-freeze strategy TBC         |
| Pan overflow<br>LSH-DX-A / LSH-DX-B / LSH-DX-C                                                              | Proposed inhibit + alarm                                           | OEM safe response                                            | Maintain safe air path                                      | Inspect drain; OEM/manual policy TBC |
| Compressor overload<br>OL-CMP-DX-A / OL-CMP-DX-B / OL-CMP-DX-C                                              | Local electrical trip                                              | OEM unit fault/changeover                                    | Isolate stopped unit                                        | Local/OEM reset; fault diagnosis     |
| DX pressure monitoring<br>PT-SUC-DX-A / PT-SUC-DX-B / PT-SUC-DX-C / PT-LIQ-DX-A / PT-LIQ-DX-B / PT-LIQ-DX-C | Monitoring only; OEM safeties govern                               | No new safety bypass                                         | No independent command                                      | OEM alarm thresholds TBC             |
| Supply static demand<br>DPT-SA-M01                                                                          | OEM cooling/staging logic                                          | Optional fan/VFD control; max TWO duty units                 | Maintain required room flows                                | Control range/station/setpoint TBC   |
| Fan proof lost<br>DPT-SF-A / DPT-SF-B / DPT-SF-C                                                            | Compressor/heat permissive inhibited via OEM                       | Stop failed unit; open/prove standby before start            | Close stopped unit to avoid reverse flow                    | Fault clear / approved changeover    |
| Filter DP high<br>PDT-F1-A / PDT-F2-A                                                                       | Alarm / OEM protection as selected                                 | No arbitrary automatic shutdown                              | Normal duty unless approved protection                      | Service filter; threshold OEM        |
| Heater airflow/high limit<br>AFS-EH-COM-01 / TSHH-EH-COM-01                                                 | DX cooling follows OEM; heater OFF independently                   | Approved ventilation mode                                    | No safety bypass                                            | Independent heater high-limit reset  |
| Standby / rotation<br>PAU-A / PAU-B / PAU-C                                                                 | Stopped unit isolated; duty pair selected                          | Max two running; prove new air path then controlled transfer | Stopped SA/RA/OA closed/proved                              | Prove normal duty after transfer     |
|                                                                                                             |                                                                    |                                                              |                                                             |                                      |

Proposal for final OEM / fire review. Factory hardwired protections remain local and cannot be bypassed by HPCP. AI22 new tags covered in the matrix. Original full-system cause/effect and275 I/O points retained. Fire is not used to initiate extraction/purge.



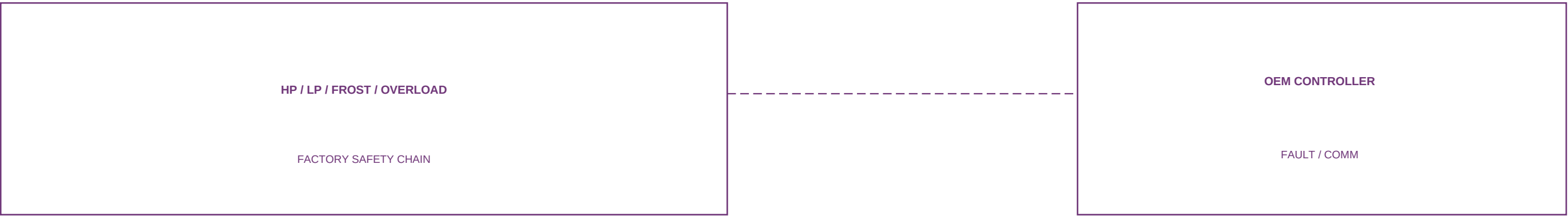
MAIN DUCT AND RETURN - DX OEM / HPCP FIELD WIRING AND I/O DEVELOPMENT

A ANALOG LOOP / OEM DATA ALTERNATIVE



Two-wire illustration only. Sensor supply, isolation and module wiring follow OEM. Route shielded twisted pair apart from motor/VFD power; shield earthing per approved panel design. Factory communication can replace extra transducers if approved.

B OEM LOCAL SAFETY / APPROVED FAULT INTERFACE



HPCP receives approved status; it does not bridge or modify factory compressor protection. Pan overflow candidate contact convention, fail-safe design and isolation require OEM review. Actual terminals and cable numbers are unassigned.

| New tag     | Category     | Factory / HPCP interface                                     | Panel assignment |
|-------------|--------------|--------------------------------------------------------------|------------------|
| PSH-DX-A    | OEM LOCAL    | Local protection; use approved factory status/communications | UNASSIGNED       |
| PSL-DX-A    | OEM LOCAL    | Local protection; use approved factory status/communications | UNASSIGNED       |
| TSSL-DX-A   | OEM LOCAL    | Local protection; use approved factory status/communications | UNASSIGNED       |
| LSH-DX-A    | DI candidate | Approved volt-free contact; contact convention TBC           | UNASSIGNED       |
| PT-SUC-DX-A | AI candidate | Approved factory data or 4-20mA isolated input               | UNASSIGNED       |
| PT-LIQ-DX-A | AI candidate | Approved factory data or 4-20mA isolated input               | UNASSIGNED       |
| OL-CMP-DX-A | OEM LOCAL    | Local protection; use approved factory status/communications | UNASSIGNED       |
| PSH-DX-B    | OEM LOCAL    | Local protection; use approved factory status/communications | UNASSIGNED       |
| PSL-DX-B    | OEM LOCAL    | Local protection; use approved factory status/communications | UNASSIGNED       |
| TSSL-DX-B   | OEM LOCAL    | Local protection; use approved factory status/communications | UNASSIGNED       |
| LSH-DX-B    | DI candidate | Approved volt-free contact; contact convention TBC           | UNASSIGNED       |
| PT-SUC-DX-B | AI candidate | Approved factory data or 4-20mA isolated input               | UNASSIGNED       |
| PT-LIQ-DX-B | AI candidate | Approved factory data or 4-20mA isolated input               | UNASSIGNED       |
| OL-CMP-DX-B | OEM LOCAL    | Local protection; use approved factory status/communications | UNASSIGNED       |
| PSH-DX-C    | OEM LOCAL    | Local protection; use approved factory status/communications | UNASSIGNED       |
| PSL-DX-C    | OEM LOCAL    | Local protection; use approved factory status/communications | UNASSIGNED       |
| TSSL-DX-C   | OEM LOCAL    | Local protection; use approved factory status/communications | UNASSIGNED       |
| LSH-DX-C    | DI candidate | Approved volt-free contact; contact convention TBC           | UNASSIGNED       |
| PT-SUC-DX-C | AI candidate | Approved factory data or 4-20mA isolated input               | UNASSIGNED       |
| PT-LIQ-DX-C | AI candidate | Approved factory data or 4-20mA isolated input               | UNASSIGNED       |
| OL-CMP-DX-C | OEM LOCAL    | Local protection; use approved factory status/communications | UNASSIGNED       |
| DPT-SA-M01  | AI candidate | Approved factory data or 4-20mA isolated input               | UNASSIGNED       |

22 candidates =7 AI +3 DI +12 OEM local protections. These are not added silently to the275 baseline I/O rows. Selected factory communications / reused fault contacts can change the final additional count. Wiring is an interface proposal, not a signed terminal schedule.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

DX OEM / HPCP FIELD WIRING AND I/O DEVELOPMENT

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-038

SCALE: NTS AT A1

DRAWING REVISION 02

08 OCT 2026

SHEET 38 OF 50



MAIN DUCT AND RETURN - DX EQUIPMENT SELECTION / ROOM DUTIES / COMMISSIONING

| Unit  | Operation                    | SA L/s  | RA L/s  | OA lower bound* | DX / fan selection                        |
|-------|------------------------------|---------|---------|-----------------|-------------------------------------------|
| PAU-A | 50% / two duty / one standby | 1848.90 | 1503.34 | 345.56          | Capacity, refrigerant, nozzle and ESP TBC |
| PAU-B | 50% / two duty / one standby | 1848.90 | 1503.34 | 345.56          | Capacity, refrigerant, nozzle and ESP TBC |
| PAU-C | 50% / two duty / one standby | 1848.90 | 1503.34 | 345.56          | Capacity, refrigerant, nozzle and ESP TBC |

ROOM AIR DUTIES / MEASURED TAB RECORD REQUIRED

| GF / room                       | SA L/s | RA L/s | EA L/s | Recorded pressure target | Acceptance / unresolved item                               |
|---------------------------------|--------|--------|--------|--------------------------|------------------------------------------------------------|
| 02 / Electrical panel           | 664.80 | 605.46 | -      | 50 Pa / unverified       | OA / leakage / comfort / field TAB and specified tolerance |
| 04 / Meeting                    | 449.20 | 389.86 | -      | 50 Pa / unverified       | OA / leakage / comfort / field TAB and specified tolerance |
| 08 / Telecom panel              | 884.60 | 825.26 | -      | 50 Pa / unverified       | OA / leakage / comfort / field TAB and specified tolerance |
| 05 / Operator console           | 777.70 | 718.36 | -      | 50 Pa / unverified       | OA / leakage / comfort / field TAB and specified tolerance |
| 06 / Reception / corridor       | 418.20 | 358.86 | -      | 50 Pa / unverified       | OA / leakage / comfort / field TAB and specified tolerance |
| 09 / Toilet group: two cubicles | 46.20  | -      | 64.89  | 25 Pa / unverified       | OA / leakage / comfort / field TAB and specified tolerance |
| 07 / Bedroom / Resting Room     | 168.20 | 108.86 | -      | 50 Pa / unverified       | OA / leakage / comfort / field TAB and specified tolerance |
| 03 / Kitchen / kitchenette      | 147.60 | -      | 147.60 | 25 Pa / unverified       | OA / leakage / comfort / field TAB and specified tolerance |
| 01 / Clean-agent cylinder room  | 141.30 | -      | TBC    | 25 Pa / unverified       | OA / leakage / comfort / field TAB and specified tolerance |

PRE-START / COMMISSIONING / PROCUREMENT HOLD

Confirm roof supports/penetrations/firestops, leakage class/pressure/gauge, weather and vapour seals; inspect fan rotation, duct cleanliness, damper end switches, drain/trap prime and every instrument scale. Prove fire stop and independent heater/DX safeties. Balance all53 terminals; record each room SA/RA/EA and pressure with sheltered outdoor reference. Confirm comfort/humidity against revised thermal loads.

\* Total makeup lower bound691.12 L/s exceeds earlier OA420.70 by270.42. Unknown CAG extract remains open; toilet net-18.69 and kitchen0 do not establish+25 Pa. The52 Rev12 pressure paths are preliminary allowances excluding bank/nozzle/OA details; they do not select a DX fan. Complete OEM coils/dirty filters, actual accessories and full fan curve before procurement.

No capacities, refrigerant safety settings, airflow tolerances, leakage measurements or certified diffuser throw/NC are invented. All room duties / necks / faces remain in the full retained registers. Installed levels and final engineering approval remain open.